



AOS-CX 10.17.xxxx Monitoring Guide For 4100i, 6000, 6100 Switch Series

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

Latest version available online

Command syntax notation conventions

About the examples

Identifying switch ports and interfaces

Applicable products

This document applies to the following products:

- HPE Aruba Networking 4100i Switch Series (JL817A, JL818A)
- HPE Aruba Networking 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A, S4R20A, S4R21A, S4R22A, S4R23A, S4R24A, S4R25A, S4R26A, S4R27A, S4R28, S4R29A)
- HPE Aruba Networking 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
example - text	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear li

Convention	Usage
	ke the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch (CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch (config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch (config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch (config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 4100i Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the HPE Aruba Networking 6000 and 6100 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

Monitoring hardware through visual observation

Subtopics

Confirming normal operation of the switch by reading LEDs

Detecting if the switch is not ready for a failover event

Finding faulted components using the switch LEDs

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.



NOTE

For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.
On the front of the switch, verify that the states of the following LEDs are On Green:
2.
 - Power
 - Health

Verify that the Health LEDs of all installed line modules are On Green.

3. Verify that the Health LEDs of all installed management modules are On Green.
4. Verify that the network ports are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:
 - On Green (normal operation)
 - Off (no line module installed)
 - b. On each line module, verify that each port LED is in one of the following states:
 - On Green, Half-Bright Green, or Flickering Green (normal operation)
 - Off (no cable connected or port off by default in config)
5. Verify that the power supplies are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:
 - On Green (normal operation)
 - Off (no power supply installed)
 - b. On each power supply, verify that LEDs are in the following states:
 - Power LED: On Green
 - Fault LED: Off
6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:
 - a. Verify that the LEDs for the fabric modules are in one of the following states:
 - On Green (normal operation)
 - Off (component not installed)
 - b. Verify that the LEDs for the fan trays and fans are On Green.
7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:
 - Health LED is On Green.
 - Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.



NOTE

Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails. For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Detect if the standby management module is shut down.
If the standby management module is shut down, the LED states are as follows:
 - The standby management module health LED is Off.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.
2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:
 - The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
 - The standby management module Booting LED is Slow Flash Green when the AOS-CX operating system is booting.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.
3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:
 - On the active management module, in the Status Rear section, the LED for the fabric module is Off.

- On the rear display module, the LED for the fabric module is Off.
- On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.



NOTE

All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.). For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the section of the .

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.
The chassis health LED is located on the front of the switch and on the rear panel of the switch.
2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:
The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.
3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:
 - The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
 - The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.
4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.
The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.

5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state: Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

IP Flow Information Export

IP Flow Information Export (IPFIX) is an embedded network flow analysis tool that compiles characteristic and measured properties of flows and sends flow reports to internal or external flow collectors. IPFIX is configurable via the command-line or REST interfaces. With IPFIX, customers configure flow records with match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports.

Compatibility with Traffic Insight

The AOS-CX traffic insight feature allows monitoring of large amount of data that it collects from various flow exporters like IPFIX, and provides the ability to filter, aggregate, and sort the data based on user flow monitor requests. Traffic insight tracks different monitor requests simultaneously and provides monitor reports per request. For more information on configuring the Traffic Insight features, refer to the AOS-CX Security Guide.

Information Elements

The IPFIX Information Elements (IE) are entities that are defined and maintained by the Internet Assigned Numbers Authority (IANA). They are characterized by a unique piece of information they can provided about a flow. Information Elements may be either private or public. Private Information Elements are exported with a Private Enterprise Number (PEN).

AOS-CX can act as an intermediate collecting process for flow reports from hardware to append certain additional IPFIX information elements to the flow reports. When configured, the software will act as an intermediate exporting process to export the augmented flow reports to any configured flow exporters.

AOS-CX supports the standard and private information elements shown in the tables below.



NOTE

Not all switches support all information elements. View a list of information elements supported by your switch using the command **show flow monitor <monitor-name> information-elements**. All Standard Information Element registration information can be found on the IANA website, at <https://www.iana.org/assignments/ipfix/>.

Standard Information Elements

octetDeltaCount

packetDeltaCount

protocolIdentifier

sourceTransportPort

sourceIPv4Address

ingressInterface

destinationTransportPort

destinationIPv4Address

vlanId

ipVersion

flowStartMicroseconds

flowEndMicroseconds

paddingOctets

ingressPhysicalInterface

About individual Information Elements

The following IEs are used for both IPv4 and IPv6 flows:

- cngSourceIP
- cngDestinationIP

They are always 16 bytes wide. The IP version of the flow is indicated in the **ipVersion** IE. For IPv4 addresses, the first 12 octets are zero and the IPv4 address is encoded in the last 4 octets. For IPv6 addresses, the full 16 bytes encode the address.

The following IEs reflect the hardware interface ID used by the associated flow:

- ingressPhysicalInterface

- egressPhysicalInterface

The hardware interface ID does not match the front-panel port number of the interface. The hardware interface ID can be mapped to the front-panel port number by using the REST API to query for **hw_intf_info** for all interfaces and saving the **switch_intf_id** value together with the associated front-panel name to form a mapping table.

Example using curl and jq:

```
bash
$ curl -X GET
    "{SWITCH}/rest/{VERSION}/system/interfaces?
attributes=hw_intf,info,name&depth=2"
-H "x-csrf-token: {CSRFTOKEN}"
-b {AUTH COOKIE FILE}
| jq 'to_entries
    | map({port: .key, switch_intf_id:
(.value.hw_intf_info.switch_intf_id | tonumber)})
    | sort_by(.switch_intf_id)
    | map({(.port): .switch_intf_id})
    | add'
{
  "1/1/6": 1,
  "1/1/2": 2,
  "1/1/1": 3,
  ...
}
```

Subtopics

Flow monitors

Flow Records

Flow Exporters

Destinations

Configuring IP Flow Information Export on 4100i, 6000, and 6100 Switches

Flow monitoring commands

Flow monitors

A flow monitor is applied to an interface to perform network traffic monitoring. A flow monitor consists of a flow record, a flow cache, and optional flow exporters. A flow record must be created and assigned to the flow monitor for the monitoring process to function. Flow data is compiled from the network traffic on the interface and stored in the flow cache based on the match (key) and collect (non-key) fields in the flow record. Data from the flow cache is exported by the flow exporters assigned to the flow monitor. 4100i,

6000, and 6100 switch series support a maximum of sixteen flow exporters that can be applied to a single flow monitor.

Flow Records

A flow record defines match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters. On the 4100i, 6000, and 6100 switch series a maximum of sixteen flow records can be created.

There are six mandatory match fields, of which the IP match fields must be of the same type (IPv4 or IPv6).



NOTE

A flow record is invalid if it does not contain one of the supported sets of match fields.

The supported sets of match fields are:

- IPv4 version
- IPV4 source address
- IPv4 destination address
- IPv4 protocol
- Transport destination port
- Transport source port
- IPv6 version
- IPv6 source address
- IPv6 destination address
- IPv6 protocol
- Transport destination port
- Transport source port

Flow Exporters

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports. 4100i, 6000, and 6100 Switch series support a maximum of sixteen flow exporters that can be applied to a single flow monitor.

Destinations

About this task

The destination specifies where flow reports are sent. There are two possible destination types for a flow exporter, but only one can be used at a time within the 4100i, 6000, and 6100 switch series:

Procedure

Hostname or IP address of a device .

Results

A flow exporter can only send flow reports to one destination. The destination type specifies which destination to use. If no destination type is specified, the default destination type is the hostname or IP address of a device supported on 4100i, 6000, and 6100 Switch series. A destination of each type can be configured, but only the one corresponding to the destination type is used. If there is no destination corresponding to the destination type, then the flow exporter configuration is incomplete. If a new destination of a particular type is configured, it will replace the destination of that type that was previously configured.

- Traffic Insight instance

Configuring IP Flow Information Export on 4100i, 6000, and 6100 Switches

The following list describes the steps required to configure a IP flow information export (IPFIX) solution:

- Step one: Create flow records
- Step two: Configure flow exporter(s)
- Step three: Configure monitor(s)

- Step four: Apply a flow monitors to interface(s)

**NOTE**

IPv6 related commands are only applicable to switches that support IPv6 protocol.

Step one: Create Flow Records

Flow Records are used to define the data that will be added to the IPFIX template. This example configures one record for IPv4 and one for IPv6.

```
switch(config)# flow record flowRecordv4
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
switch(config-flow-record)# collect application https url
switch(config)# flow record flowRecordv6
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect timestamp absolute first
```

Next, use the **show flow record** command to verify the configuration.

Step two: Configure flow exporter(s)

In this step, you can define an exporter to send to an external destination by hostname or IP address, or to an internal destination such as Traffic Insight. The example below configures IPFIX to export data to an external address/hostname:

```
switch(config)# flow exporter flowExternal
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 11.1.1.1
switch(config-flow-exporter)# show flow exporter
```

```

-----
----
Flow exporter 'flowExternal'
-----
----
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Hostname or IP address
Destination            : 11.1.1.1
Transport Configuration
Protocol               : udp
Port                   : 4739

```

To configure IPFIX to export to Traffic Insight, first configure Traffic Insight.

```

switch(config)# traffic-insight TI
switch(config-ti-TI)# source ipfix
switch(config-ti-TI)# monitor topN type topN-flows
switch(config-ti-TI)# monitor appFlow type application-flows
switch(config-ti-TI)# enable

```

Next, configure the flow exporter for Traffic Insight

```

switch(config)# flow exporter flowExpTI
switch(config-flow-exporter)# export-protocol ipfix
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight TI

```

You can use the **show flow exporter** command to verify the flow exporter configuration for Traffic Insight

```

switch(config)# show flow exporter flowExpTI
-----
----
Flow exporter 'flowExpTI'
-----
----
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Traffic Insight
Destination            : TI
Transport Configuration
Protocol               : udp
Port                   : 4739

```

Finally, use the **show run traffic-insight** command to verify the Traffic Insight configuration:

```

switch(config)# show running-config traffic-insight
traffic-insight TI
enable
source ipfix

```

```
!  
monitor topN type topN-flows entries 5  
monitor appFlow type application-flows
```

Step three: Configure the monitor(s)

First, configure an IPv4 flow monitor.

```
switch(config)# flow monitor flowMonv4  
switch(config-flow-monitor)# record flowRecordv4  
Switch (config-flow-monitor)# exporter flowExternal  
switch(config-flow-monitor)# exit
```

Next, configure an IPv6 flow monitor.

```
switch(config)# flow monitor flowMonv6  
switch(config-flow-monitor)# record flowRecordv6  
switch(config-flow-monitor)# exporter flowExternal  
switch(config-flow-monitor)# exit
```

Once both flow monitors are created, use the **show flow monitor** command to verify the flow monitor configurations.

```
switch(config-flow-monitor)# show flow monitor
```

```
-----  
----  
Flow monitor 'flowMonv4'  
-----  
----  
Status                : Accepted  
Flow Record           : flowRecordv4  
Flow Exporter(s)      : flowExternal  
Cache Configuration  
Inactive Timeout      : 120  
Active Timeout        : 1800  
-----  
----  
Flow monitor 'flowMonv6'  
-----  
----  
Status                : Accepted  
Flow Record           : flowRecordv6  
Flow Exporter(s)      : flowExternal  
Cache Configuration  
Inactive Timeout      : 120  
Active Timeout        : 1800
```

Flow monitoring commands

Subtopics

- [exporter](#)
- [flow exporter](#)
- [flow monitor](#)
- [flow record](#)
- [ipfix flow monitor \(interface\)](#)
- [show flow exporter](#)
- [show flow monitor](#)
- [show flow record](#)
- [show ipfix flow-table-utilization](#)

exporter

Syntax

```
exporter <EXPORTER-NAME>
no exporter <EXPORTER-NAME>
```

Description

Assigns a flow exporter to **flow monitor** in the **config-flow-monitor** context.

The **no** form of this command removes the flow exporter from the selected monitor.

Parameter	Description
<code><EXPORTER-NAME></code>	Specifies the name of the flow exporter assigned to the monitor.



NOTE

A maximum of sixteen flow exporters can be applied to each flow monitor.

Examples

Assigning flow exporter, **flow-exporter-1**, to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# exporter flow-exporter-1
```

Removing a flow exporter assigned to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# no exporter flow-exporter-1
```

Attempting to assign non-existent flow exporter, **flow-exporter-5**, to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# exporter flow-exporter-5
Flow exporter 'flow-exporter-5' does not exist.
switch(config-flow-monitor)#
```

Attempting to assign flow exporter, **flow-exporter-6**, to monitor, **flow-monitor-3**, when two exporters are assigned:

```
switch(config)# flow monitor flow-monitor-3
switch(config-flow-monitor)# exporter flow-exporter 4
switch(config-flow-monitor)# exporter flow-exporter 5
switch(config-flow-monitor)# exporter flow-exporter 6

switch(config-flow-monitor)#
```

Command History

Release	Modification
10.17.1000	Command introduced
10.16	Command introduced

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-flow-monitor	Administrators or local user group members with execution rights for this command.

flow exporter

Syntax

```
flow exporter <name>
```



```

description <description>
destination
traffic-insight}
traffic-insight <instance-name>
no ...
transport udp <port>

```

Description

A flow exporter is the part of the IP Flow Information Export (IPFIX) feature that defines how a flow monitor exports flow reports. You can assign the same flow exporter configuration to more than one flow monitor. Each flow exporter includes a destination setting that identifies the device to which the flow reports are sent

Parameter	Description
<name>	Name of the flow exporter, up to 64 characters.
export-protocol ipfix	Define an export protocol for the flow exporter. The default ipfix protocol is the only protocol currently available.
description <description>	A description of the flow exporter, up to 256 characters and spaces.
destination	Configure the export destination
<hostname>	The exporter sends flow records to the specified hostname destination. The hostname can be a string of up to 64 characters.
<ip6addr>	The exporter sends flow records to this IPv6 address destination.
vrf <vrfname>	You can optionally include the name of the destination VRF in the destination definition on 5420, 6200, 6300, 6400, 8100, 8360, 9300S Switch series.
type	Configure the type of the destination.
hostname-or-ip-addr	Define the destination type as a hostname or IP address.
traffic-insight <name>	Define the destination type as a traffic insight instance.
no ...	Negate any configured parameter.
traffic-insight <INSTANCE-NAME>	Specify the a Traffic Insight instance to be used as the destination.
template data timeout <timeout>	A flow exporter template describes the format of exported flow reports. Therefore, flow reports cannot be decoded properly without the corresponding templates. This setting defines how o

Parameter	Description
	ften the flow exporter will resend templates to the flow monitor. The supported range is 1 - 86400 seconds, and the default is 600 seconds.
<code>transport udp <port></code>	Transport protocol and port for sending flow record reports. The default port is port 4739.

Usage

The following table shows the maximum supported of flow monitors and flow exporters for each switch model.

Examples

The following example creates a flow exporter configuration named **exporter-1**:

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight-instance-1
```

The following example attempts to create more than the maximum number of allowed flow exporters:

```
switch(config)# flow exporter exporter-1
switch(config)# flow exporter exporter-2
<--OUTPUT OMITTED FOR BREVITY-->
switch(config)# flow exporter exporter-16
switch(config)# flow exporter exporter-17
Unable to add another flow exporter; the limit is 16.
```

The following example configures the destination type:

```
switch(config)# destination type {hostname-or-ip-addr | traffic-insight}
switch(config)# destination {<HOSTNAME> | <IPV4-ADDR> | <IPV6-ADDR>}
switch(config)# destination traffic-insight <INSTANCE-NAME>
```

Command History

Release	Modification
10.17.1000	Command introduced on 4100i, 6000, and 6100 Switch series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config config-flow- exporter	Administrators or local user group members with execution rights for this command.

flow monitor

Syntax

```
flow monitor <name>  
    exporter <name>  
cache timeout {inactive <timeout>}|{active <timeout>  
    description <description>  
    record <name>
```

Description

On HPE Aruba Networking 4100i, 6000, and 6100 Switch series, a flow monitor is the part of the IP Flow Information Export (IPFIX) feature that performs network monitoring for the selected interface. A flow monitor configuration consists of a flow record, a flow cache, and one or more associated flow exporters. A flow monitor compiles data from the network traffic on the interface and stores it in the flow cache in a format defined by the flow record. The flow exporters associated with the monitor then export data from the flow cache to the flow exporter destination.



NOTE

HPE Aruba Networking 4100i, 6000, and 6100 Switch series support a maximum of sixteen flow exporters that can be applied to a single flow monitor. If no software augmentation of flows is required, there is no need to configure a flow collector or flow monitor.

Parameter	Description
<name>	Name of the flow monitor, up to 64 characters.
cache timeout active <timeout>	Use the cache timeout parameter to define an active or inactive timeout for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer than the inactive timeout.

Parameter	Description
	The active timeout range is 120 - 604800. The default active timeout value is 1800 and inactive timeout value is 120.
cache timeout inactive <timeout>	Use the cache timeout parameter to define an inactive timeout for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer than the inactive timeout. For 4100i, 6000, and 6100 Switch Series, the minimum supported inactive timeout range is 120 seconds. The maximum inactive timeout range is 604800 seconds, and the default inactive timeout is 120 seconds.
description	A description up to 256 characters long, including spaces.
exporter <name>	Assign a flow exporter to a flow monitor.
record <name>	(For HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360, 9300, 9300S, 10040 Switch series) Assigns a flow record to a flow monitor.

Examples

The following example creates a flow monitor configuration named **monitor-1**.

```
switch(config)# flow monitor monitor-1
switch(config-flow-monitor)# description Monitor for analyzing basic ipv4
traffic
switch(config-flow-monitor)# exporter flow-exporter-1

switch(config-flow-monitor)# record flow-record-1
switch(config-flow-monitor)# cache timeout inactive 120
switch(config-flow-monitor)# cache timeout active 1500
```

The following example assigns **collector-1** as the collector associated with this monitor.

```
switch(config)# flow monitor monitor-1
switch(config-flow-monitor)# collector collector-1
```

The following workflow changes the flow record assigned to a flow monitor.

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# record flow-record-2
```

Create more than the maximum number of allowed flow monitors

```
switch(config)# flow monitor monitor-1
switch(config)# flow monitor monitor-2
<--OUTPUT OMITTED FOR BREVITY-->
switch(config)# flow monitor monitor-16
```

```
switch(config)# flow monitor monitor-17
```

No more than 16 flow monitors can be configured. Another flow monitor must be removed first.

Add or modify the description of flow exporter **flow-exporter-1**

```
switch(config)# flow exporter flow-exporter-1
switch(config-flow-exporter)# description Exports flows to 10.2.3.45:2055
switch(config-flow-exporter)# description Exports flows to Traffic Insight
```

Command History

Release	Modification
10.17.1000	Command introduced on the 4100i, 6000, and 6100 switch series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

flow record

Syntax

```
flow record <record-name>
  match
ip {source|destination} address
ip protocol|version
ipv6 {source|destination} address
ipv6 protocol|version
transport {source | destination} port
  collect
counter {packets|bytes}
timestamp absolute last
  descriptionN <description>
```

Description

Creates or modifies a flow record and switches to the **config-flow-record** context for the flow record. Define data to be included in a flow record by configuring flow record match and collect fields.

A flow record defines match (key) fields and collection (non-key) fields. Customers configure flow records with **match** (key) fields and **collect** (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collect fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

Traffic with matching attributes (for example, traffic coming from the same interface, sent to the same destination with the same protocol) are classified as a single flow. Information for some or all of the matched settings can be collected and exported to a destination defined by the flow exporter assigned to the flow monitor.



NOTE

Traffic must match a match rule definition before it can be collected and sent. You cannot collect and send data that is not matched.



NOTE

For 4100i, 6000, and 6100 Switch series, a maximum of sixteen flow records can be created.

Parameter	Description
<code><record-name></code>	Name of the flow monitor, up to 64 characters.
<code>match</code>	match traffic according to one or more of the following key attributes: • ip source: Match traffic from the same IPv4 source. • ip destination: Match traffic to the same IPv4 destination. • ip protocol: Match traffic using the same IP version • ip version: Match traffic using the same IP protocol • ipv6 source: Match traffic from an IPv6 source. • ipv6 destination: Match traffic to an IPv6 destination. • ipv6 protocol: Match traffic using the same IPv6 version • ipv6 version: Match traffic using the same IPv6 protocol • transport {source destination} port: Match traffic by source or destination transport port
<code>description</code>	A description for the flow record up to 256 characters long, including spaces.
<code>collect</code>	Configures data fields to be included a flow record.

Parameter	Description
	• counter bytes: Collect counter data for bytes in the flow. • counter packets: Collect counter data for packets in the flow. • timestamp absolute first: Collect absolute timestamp of the first packet observed. • timestamp absolute last: Collect absolute timestamp of the last packet observed.

Examples

Adding counter and timestamp collect fields to **flow-record-1**:

```
switch(config)# flow record flow-record-1
```

Creating more than the maximum number of allowed flow records:

```
switch(config)# flow record record-1
switch(config)# flow record record-1
switch(config)# flow record record-2
<--OUTPUT OMITTED FOR BREVITY-->
switch(config)# flow record record-16
switch(config)# flow record record-17
Unable to add another flow record; the limit is 16.
```

Adding IPv4 match fields to flow record **flow-record-1** using the **ip** keyword:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
```

Adding IPv6 match fields to flow record **flow-record-2**:

```
switch(config)# flow record flow-record-2
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 version
```

Removing the IPv4 destination address match field from flow record **flow-record-1** using the **ip** keyword:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ip destination address
```

Removing the IPv4 destination address match field from flow record **flow-record-1** using the **ipv4** keyword:

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ipv4 destination address
```

Removing the transport destination port match field from flow record **flow-record-1**:

```
switch(config)# flow record flow-record-1  
switch(config-flow-record)# no match transport destination port
```

Command History

Release	Modification
10.17.1000	Command introduced on 4100i, 6000, and 6100 Switch series.
10.13	Added application https url and dns response - code parameters.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config config-flow- record	Administrators or local user group members with execution rights for this command.

ip|ipv6 flow monitor (interface)

Syntax

```
[no] ip|ipv6 flow monitor (interface)
```

Description

Enable flow monitoring on inbound and outbound interfaces by assigning a flow monitor to that interface. Only physical interfaces and LAG interfaces can be monitored. A flow monitor cannot be applied to an interface that is part of a LAG. If an unsupported application is attempted, an error message will be displayed. If the flow monitor is associated with a flow record that contains application fields as collect fields, then Application Recognition should be enabled on the same interface.

The **[no]** form of command disables the flow monitoring.



NOTE

The 4100i, 6000, and 6100 Switch Series support ingress-only interfaces; therefore, flow monitoring on these platforms is limited to inbound traffic.

Examples

Associate a flow monitor configuration named **flow-monitor-3** for IPv6 traffic on a physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ip flow monitor flow-monitor-3 in
```

Enable an IPv6 flow monitor on a physical interface

```
switch(config)# interface lag 1
switch(config-if)# ipv6 flow monitor flow-monitor-1 in
```

Command History

Release	Modification
10.17.1000	Command introduced on 4100i, 6000, and 6100 Switch series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

show flow exporter

Syntax

```
show flow exporter [<name>]
[statistics]
```

Description

Displays flow exporter statistics, configuration and status. When no exporter name is specified, the output of this command displays information for all flow exporters.

The output of this command can indicate the following status types:

- Rejected (Destination type is hostname or IP address, but no destination is specified)

- Rejected (Destination type is hostname or IP address, but the specified hostname or IP address is invalid)
- Rejected (Destination type is Traffic Insight, but no destination is specified)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance does not exist)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance is not enabled)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance source is not IPFIX)
- Rejected (Internal error: destination type is Traffic Insight, but the specified Traffic Insight instance is invalid)

Parameter	Description
<name>	Name of the flow exporter.
statistics	Adds statistical information about the flow exporter to the output.

Examples

Display the configuration of a flow exporter named **exporter-1**.

```
switch# show flow exporter exporter-1
-----
----
Flow exporter 'exporter-1'
-----
----
Description           : Exports to the first collector
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Hostname or IP address
Destination            : 192.168.0.1
Transport Configuration
    Protocol           : UDP
    Port                : 9995
```

Display statistics information for all flow exporters

```
switch# show flow exporter statistics
-----
----
Flow exporter 'exporter-1'
-----
```

```
-----
Reports sent           : 14961
-----
```

```
-----
Flow exporter 'exporter-2'
-----
```

```
-----
Reports sent           : 5
-----
```

Display information with no flow exporters configured

```
switch# show flow exporter
No flow exporters configured.
```

Display a flow exporter information with TI as a destination

```
switch# show flow exporter exporter-5
-----
Exporter Name          : exporter-5
-----
Description             : Exporter configured with TI as the destination
Status                  : Rejected (Destination type is Traffic Insight,
but the specified Traffic Insight instance does not exist)
Export Protocol         : ipfix
Destination Type        : Traffic Insight
Destination             : instance-1
Transport Configuration
Protocol                : UDP
Port                   : 2055
```

Display information for a flow exporter that has multiple destinations of different types configured and **hostname-or-ip-addr** is specified as the destination type to use.

```
switch# show flow exporter exporter-6
-----
Flow exporter 'exporter-6'
-----
Description             : TI and hostname configured, but type is hostname-
or-ip-addr
Status                  : Accepted
Export Protocol         : ipfix
Destination Type        : Hostname or IP address
Destination             : collector-1
Destination VRF         : mgmt
```

```
Transport Configuration
Protocol           : UDP
Port               : 4821
```

Command History

Release	Modification
10.17.1000	Command introduced on 4100i, 6000, and 6100 Switch series

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

show flow monitor

Syntax

```
show flow monitor
[statistics | information-elements]

show flow monitor <MONITOR-NAME>
[statistics]
```

Description

Displays flow monitor configuration and status. When no monitor name is specified, the output of this command displays information for all flow monitors.

Parameter	Description
<name>	Name of the flow monitor.

Parameter	Description
<code>statistics</code>	Includes the statistics parameter to display additional flow and cache statistics.
<code>information-elements</code>	Displays the configured information elements for the flow monitor.

Usage

The output of this command can indicate the following status types:

- Possible status types for all switches
 - Accepted
 - Rejected (Internal error: monitor does not exist)
 - Rejected (The state of one or more of the assigned flow exporters is rejected)

The possible statistics for a flow monitor are:

Examples

Display information for flow monitor **monitor-1** on 4100i, 6000, and 6100 Switch series:

```
switch# show flow monitor 'monitor-1'
-----
----
Flow monitor 'monitor-1'
-----
----
Description           : Used for IPv4 traffic analysis
Status                : Accepted
Flow Record           : record-1
Flow Exporter(s)      : exporter-1, exporter-2
Cache Configuration
Inactive Timeout      : 1800
Active Timeout        : 300
-----
----
```

Display information for flow monitor **monitor-2** on 4100i, 6000, and 6100 Switch series:

```
switch# show flow monitor 'monitor-2'
-----
----
Flow monitor 'monitor-2'
```

```

-----
----
Description          : Used for IPv6 traffic analysis
Status               : Rejected (The state of one or more of the
specified flow exporters is rejected)
Flow Record          : record-2
Flow Exporter(s)     : exporter-1
Cache Configuration
Inactive Timeout     : 1800
Active Timeout       : 2400
-----
----

```

Display information and statistics for a flow monitor on 4100i, 6000, and 6100 Switch series:

```

show flow monitor statistics
-----
----
Flow monitor 'monitor-1'
-----
----
Current Entries      : 2
Flows Added          : 6
Total Flows Terminated : 4
  Flows Aged         : 2
    Active Timeout    : 1
    Inactive Timeout  : 1
  End of Flow Detected : 2
  Forced End          : 0
  Flows Aged          : 4

```

Command History

Release	Modification
10.17.1000	Command introduced on 4100i, 6000, and 6100 Switch series.

Command Information

Platforms	Command context	Authority
4100i 6000	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

show flow record

Syntax

```
show flow record [<name>]
```

Description

Display flow record configuration and status. When no record name is specified, the output of this command displays information for all flow records.

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: failed to process record)
- Rejected (Mix of IPv4 and IPv6 match fields is not allowed. Specify match fields of the same IP version (IPv4 or IPv6))
- Rejected (Incomplete match fields. The mandatory match fields are: version, source address, destination address protocol, transport destination port, and transport source port)

Parameter	Description
<name>	Name of the flow record.

Examples



NOTE

IPv6 related commands are only applicable to switches that support IPv6 protocol.

Display the configuration of a flow record named **flow-record-1**.

```
switch# show flow record record-1
```

```
-----  
----
```

```
Flow record  'record-1'
```

```

-----
----
Description          : Used for IPv4 traffic analysis
Status               : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    transport destination port
    transport source port
Collect Fields
counter bytes
    counter packets

```

Display the information of a specific flow record.

```

switch# show flow record record-1
-----
----
Flow record  'record-1'
-----
----
Description          : Used for IPv4 traffic analysis
Status               : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address

    transport destination port
    transport source port
Collect Fields
    counter bytes
    counter packets

```

Display information for all flow records

```

switch# show flow record
-----
----
Flow record  'record-1'
-----
----
Description          : Used for IPv4 traffic analysis

```



```
Status : Accepted
```

```
Match Fields
```

```
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    transport destination port
    transport source port
```

```
Collect Fields
```

```
    counter bytes
    counter packets
```

```
-----
----
Flow record  'record-2'
```

```
-----
----
Description : Used for IPv6 traffic analysis
```

```
Status : Accepted
```

```
Match Fields
```

```
    ipv6 destination address
    ipv6 protocol
    ipv6 source address
    ipv6 version
    transport destination port
    transport source port
```

```
Collect Fields
```

```
    application name
    counter bytes
    counter packets
```

```
...
```

Display information for a specific flow record

```
switch# show flow record record-3
```

```
-----
----
Flow record  'record-3'
```

```
-----
----
Description : Used for IPv4 traffic analysis
```

```
destination address, protocol, transport destination port, and transport
source port.)
```

```
Match Fields
```

```
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
```

```
Collect Fields
counter bytes
counter packets
```

Display information with no flow records configured

```
switch# show flow record
No flow records configured
```

Command History

Release	Modification
10.17.1000	Command introduced on the 4100i, 6000, and 6100 Switch series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

show ipfix flow-table-utilization

Syntax

```
show ipfix flow-table-utilization
```

Description

Displays the current IPFIX flow table utilization per VSF member, including both the maximum capacity and the current number of current flows.

Examples

Displaying current IPFIX flow table utilization:

```
switch# show ipfix flow-table-utilization
=====
Slot name | flow resources capacity | flow resources in use
-----|-----|-----
```

1/1	3072	1245
1/3	3072	980

Command History

Release	Modification
10.17.1000	Command introduced on 4100i, 6000, and 6100 Switch series.

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Manager (#)	Administrators or local user group members with execution rights for this command.

Alarms

Alarms allow you to configure physical alarms for alarm input ports and specific systems events.



CAUTION

Alarms are only available on the 4100i switch.

Subtopics

Alarm commands

Alarm commands

Subtopics

alarm

alarm input

alarm snooze

show alarm

alarm

Syntax

```
alarm
    power-supply|temperature
action <LOG-AND-TRAP>|<RELAY>
    input IN1|IN2
action <LOG-AND-TRAP>|<RELAY>
name <STRING>
trigger <CLOSED>| <OPEN>
    no...
    snooze
<time in minutes>
repeat
```

Description

Configures input alarm and actions, or global events to be forwarded to the output alarm port, or temporarily disables (snoozes) active event forwarding to the alarm relay for a specified time interval.

Parameter	Description
temperature	Selects the alarm for ambient temperatures reaching the threshold limit of 70°C.
power-supply	Selects the alarm events from the power supply.
action	Specifies the action to be taken when the monitored event occurs.
<LOG-AND-TRAP>	Generates an event log and SNMP trap.
<RELAY>	Relays an event to alarm output port.
input IN1 IN2	Specifies the input alarm port.

Parameter	Description
<code>action</code>	Specifies the action to be taken when the monitored event occurs.
<code><LOG-AND-TRAP></code>	Generates an event log and SNMP trap.
<code><RELAY></code>	Relays an event to alarm output port.
<code>name <STRING></code>	Specifies the external device connected.
<code>trigger</code>	Triggers an alarm based on a normally open or closed circuit.
<code><CLOSED></code>	Generates an alarm event when the circuit is closed.
<code><OPEN></code>	Generates an alarm event when the circuit is open.
<code>no ...</code>	Negates any configured parameter or removes the specified configuration.
<code>snooze</code>	Snooze the relay action for specified time interval.
<code><time in minutes></code>	Specifies the snooze time in minutes. The range is 0 - 1440 minutes..
<code>repeat</code>	Repeats the previous snooze time.

Examples

Disabling a temperature event to remove the configuration for all actions associated with the event:

```
switch(config)# no alarm temperature
```

Configuring an alarm for a temperature event for the log-and-trap action:

```
switch(config) # alarm temperature action log-and-trap
```

Removing the configuration for the temperature event for a log-and-trap action:

```
switch(config) # no alarm temperature action log-and-trap
```

Configuring an alarm for a power-supply event for the relay action:

```
switch(config) # alarm power-supply action relay
```

Removing the configuration for the power-supply event for the relay action:

```
switch(config) # no alarm power-supply action relay
```

Configuring an alarm on input port 1 named Door-Sensor:

```
switch(config) # alarm input IN1 name Door-Sensor
```

Removing the configuring for an alarm on input port 1:

```
switch(config) # no alarm input IN1
```

Configuring an alarm on input port 1 with log-and-trap action:

```
switch(config) # alarm input IN1 action log-and-trap
```

Removing the configuration for an alarm on input port 1 with log-and-trap action:

```
switch(config) # no alarm input IN1 action log-and-trap
```

Configuring an alarm on input port 1 to trigger an alarm when the door sensor circuit is closed:

```
switch(config) # alarm input IN1 trigger closed
```

Configuring an alarm relay action for 10 minutes:

```
switch(config) # alarm snooze 10
```

Configuring an alarm relay action to be repeated with previously configured snooze time:

```
switch(config) # alarm snooze repeat
```

Command History

Release	Modification
10.08	Feature introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

alarm input

Syntax

```
alarm input {IN1 | IN2} [name <STRING>] [action <LOG-AND-TRAP> | <RELAY>]
[trigger <CLOSED | OPEN>]
no alarm input {IN1 | IN2} [name <STRING>] [action <LOG-AND-TRAP> | <RELAY>]
[trigger <CLOSED | OPEN>]
```

Description

Configures input alarm and actions. The **no** form of this command removes the specified configuration.

Parameter	Description
{IN1 IN2}	Specifies the input alarm port..
name	Specifies the external device connected.
<STRING>	Descriptive string.
action	Specifies the action to be taken when the monitored event occurs.
<LOG-AND-TRAP>	Generates an event log and SNMP trap.
<RELAY>	Relays an event to alarm output port.

Parameter	Description
trigger	Triggers an alarm based on a normally open or closed circuit.
<CLOSED>	Generates an alarm event when the circuit is closed.
<OPEN>	Generates an alarm event when the circuit is open.

Examples

Configuring an alarm on input port 1 named Door-Sensor:

```
switch(config)# alarm input IN1 name Door-Sensor
```

Removing the configuring for an alarm on input port 1:

```
switch(config)# no alarm input IN1
```

Configuring an alarm on input port 1 with log-and-trap action:

```
switch(config)# alarm input IN1 action log-and-trap
```

Removing the configuration for an alarm on input port 1 with log-and-trap action:

```
switch(config)# no alarm input IN1 action log-and-trap
```

Configuring an alarm on input port 1 to trigger an alarm when the door sensor circuit is closed:

```
switch(config)# alarm input IN1 trigger closed
```

Command History

Release	Modification
10.08	Feature introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

alarm snooze

Syntax

```
alarm snooze [time in minutes] [repeat]
```

Description

Configures any active event forwarded to alarm relay to be snoozed.

Parameter	Description
time in minutes	Specifies the value for time in minutes. The range is 0 - 1440 minutes..
repeat	Repeats the previous snooze time.

Examples

Configuring an alarm relay action for 10 minutes:

```
switch(config)# alarm snooze 10
```

Configuring an alarm relay action to be repeated with previously configured snooze time:

```
switch(config)# alarm snooze repeat
```

Command History

Release	Modification
10.08	Feature introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

show alarm

Syntax

```
show alarm
  input [IN1 | IN2]
  power supply
  temperature
  timer
```

Description

Shows all of the details of global status monitoring alarm events.

Parameter	Description
{IN1 IN2}	Specifies the input alarm port..
power supply	Selects the alarm events from the power supply.
timer	Shows the status of an alarm snooze timer's status and duration .
temperature	Selects the alarm for ambient temperatures reaching threshold limits.

Examples

Showing details for all global alarm events on the switch:

```
switch# show alarm
Alarm Snooze Timer Status: active
Duration remaining: 1 min 32 sec
-----
Global Alarm: Temperature
```

Alarm Event	Status	log-and-trap	Relay
Temperature	inactive	true	false
Global Alarm: power-supply			
Alarm Event	Status	log-and-trap	Relay
power-supply	active	false	true

Showing details for the temperature alarm:

```
switch# show alarm temperature
Alarm Snooze Timer Status: Active
Duration remaining: 5 min 36 sec
```

Global Alarm: Temperature

Alarm Event	Status	log-and-trap	Relay
Temperature	inactive	true	false

Showing the status of an alarm snooze timer when it is inactive :

```
switch# show alarm timer
Alarm Snooze Timer Status: inactive
```

Showing the status of an active 3-minute alarm snooze timer:

```
switch# show alarm timer
Alarm Snooze Timer Status: active
Duration remaining: 2 min 55 sec
```

Showing details for all alarm input ports on the switch:

switch# show alarm input				
Alarm Snooze Timer Status: inactive				
Input Alarm IN1, Name: Door-Sensor				
Alarm Port	Status	log-and-trap	Relay	Trigger
IN1	inactive	true	false	closed
Input Alarm IN2, Name: N/A				
Alarm Port	Status	log-and-trap	Relay	Trigger
IN2	active	false	true	open

Showing details for alarm input ports IN1:

```
switch# show alarm input IN1
Alarm Snooze Timer Status: inactive
```

Input Alarm IN1, Name: Door-Sensor

Alarm Port	Status	log-and-trap	Relay	Trigger
IN1	inactive	true	false	closed

Command History

Release	Modification
10.08	Feature introduced.

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution rights for this command.

Boot commands

Subtopics

[Boot commands](#)

Boot commands

Subtopics

[boot set-default](#)

[boot system](#)

[show boot-history](#)

boot set-default

Syntax

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```



NOTE

For more information on features that use this command, refer to the Fundamentals Guide or the Monitoring Guide for your switch model.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

Syntax

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
primary	Selects the primary operating system image for this reboot and sets the configured default operating system image to primary for future reboots.
secondary	Selects the secondary operating system image for this reboot and sets the configured default operating system image to secondary for future reboots.
serviceos	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovers OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.
You can use the **boot set-default** command to change the configured default operating system image.
- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system
Do you want to save the current configuration (y/n)? n
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

Syntax

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.



NOTE

To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
all	Optional. Shows boot information for the active management module.
vsf member <1-10>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system - generated 128 - bit string.
Current Boot, up for <time>	For the current boot, the show boot - history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot - history command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: • <DAEMON - NAME> crash: The daemon identified by <DAEMON - NAME> caused the module to boot. • Kernel crash: The operating system software associated with the module caused the module to boot. • Uncontrolled reboot: The reason for the reboot is not known. • Reboot requested through database: The reboot occurred because of a request made through the CLI or other API.

Table 1. Description of reboots handled through the database

Boot History String	Description
Reboot requested by user	A user requested a switch reboot through the CLI or web UI.
Reset button pressed	The switch detected a short - press of the reset button
Backplane fault	A backplane fault occurred.
Configuration change	A configuration change resulted in a reboot.
Configuration version migration	A configuration version migration occurred which required a reboot.
Console error	The console failed to start.
Fabric fault	A fabric fault occurred.

Boot History String	Description
All line modules faulted	A zero line card condition occurred.
Redundancy switchover requested	A user requested a redundancy switchover.
Redundant Management communication timeout	The standby management module has taken over from an unresponsive active management module.
Redundant Management election timeout	A failure to elect a standby management module in the allotted time.
Critical service fault (error)	A daemon critical to switch operation has stopped functioning. An extra error string may be present to describe the error in detail.
VSF autojoin renumber	Reset triggered by VSF autojoin.
VSF member renumbered	A user requested a renumber of a VSF member.
VSF switchover requested	A user requested a VSF switchover.
VSX software update	Reset triggered by a VSX software update.
Chassis critical temperature	Chassis operating temperature exceeded.
Chassis low critical temperature	Chassis temperature below the minimum operating threshold.
Chassis insufficient fans	Insufficient fans to cool the chassis.
Chassis unsupported PSUs/fans	Unsupported or misconfigured PSUs or system fans.
Management module critical temperature	Management module operating temperature exceeded.
ISSU SMM update	Standby management module reboot triggered by an In - Service Software Upgrade (ISSU).
ISSU switchover	Redundancy switchover triggered by an In - Service Software Upgrade.
ISSU aborted	Standby management module reset triggered by failure during an In - Service Software Upgrade.
Rollback timer expired	Reset triggered by the ISSU rollback timer expiring.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====
Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs
Index : 1
Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version:
FL.10.14.0000-1619-ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2
Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1
Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in
PSU 1/1
switch#
Management module
=====
Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
```

```

10 Aug 17 12:45:46 : dune_agent crashed
...
Management module
=====
Index : 3
Boot ID : flbf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs
Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database
Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database
Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
switch# show boot-history
Management module
=====
Index : 2
Boot ID : a6lad00d10864c748bc7893a5d4af2e4
15 Dec 23 19:02:02 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:02:02 : Switch boot count is 0
15 Dec 23 19:02:17 : PSU 1/1: Fault detected
Index : 1
Boot ID : 30d831bbfdfa425baf50a629ee01b185
15 Dec 23 19:01:58 : Power on reset with 0x1, Version: FL.10.13.1000AF
15 Dec 23 19:01:58 : Switch boot count is 0
switch# show boot-history vsf member 2
Member-2
=====
Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs
Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

L1-100Mbps downshift

The speed downshift feature allows the user to link-up at sub-optimal speeds when failing to link-up at the highest advertised speed. There are fixed number of link attempts made to establish link at highest advertised speed and when all of them fail and attempt is made to link-up at a lower possible speed.

This feature requires underlying PHY to have support for the same and hence capability is only added to select set of ports. If a link cannot be established at the highest common denominator within a set number of link attempts, the PHY advertises the next highest speed using auto-negotiation.

Subtopics

Limitations with speed downshift

L1-100Mbps downshift commands

Limitations with speed downshift

- Link up may be delayed as certain number of retries are done to establish the link at highest advertise speeds by both link partners before downshifting.
- Link may be established at sub-optimal speed.

L1-100Mbps downshift commands

Subtopics

downshift enable

show interface

show interface downshift-enable

show running-config interface

downshift enable

Syntax

```
downshift-enable  
no downshift-enable
```

Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

Examples

```
switch(config-if) # interface 1/1/1  
switch(config-if) # downshift-enable  
Warning: this is a non-standard mode for use only when standards-based  
auto-negotiation is not able to establish a stable link. Enabling this  
may cause the port to link at a lower than expected speed and should  
not be used on ports that are members of a LAG. Support calls may require  
this feature to be disabled  
Continue (y/n)?  
switch(config-if) #
```

When automatic downshift is enabled:

```
switch(config-if) # show running-config interface  
interface 1/1/1  
    downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if) # interface 1/1/1  
switch(config-if) # no downshift-enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

show interface

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-readable]]
show interface [<IFNNAME>] monitor [human-readable]
show interface [lag | vlan ] [<ID>] [brief]
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
	Specifies the port identifier range.

Parameter	Description
<IFRANGE>	
brief	Shows brief info in tabular format.
physical	Shows the physical connection info in tabular format.
extended	Shows additional statistics, including the tx filtered and rx filtered counters. - Rx filter packets are protocol packets received when the protocol is disabled on the switch and there is only one port in the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LLDP. - An example of a Tx filtered packet would be a multicast packet being filtered from going out of the ingress port.
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G. This is available only in the CLI interface output.
non-zero	Shows only non zero statistics.
LAG	Shows LAG interface information.
monitor	Continuously monitor interface statistics.
VLAN	Shows VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1 - 256
<VLAN-ID>	Specifies the VLAN ID. Range: 1 - 4094

Examples

```
switch# show interface 1/1/1
Interface 1/1/1 is up
Admin state is up
Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
```



```

Link transitions: 1
Description: backup data center link
Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 1GbT
Full-duplex
qos trust none
Speed 1000 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off

                Energy-Efficient Ethernet is enabled
                MDI mode: MDIX

```

```

L3 Counters: Rx Enabled, Tx Enabled
Rate collection interval: 300 seconds

```

Rates	RX	TX	Total (RX+TX)
-----	-----	-----	-----
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistics	RX	TX	Total
-----	-----	-----	-----
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
L3 Packets	0	0	0
L3 Bytes	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0
Other	0	0	0

Showing information when interface 1/1/1 is configured:

MDI mode: MDIX

VLAN Mode: native-untagged

Native VLAN: 1

Allowed VLAN List: all

Rate collection interval: 300 seconds

Rates	RX	TX	Total (RX+TX)
-----	-----	-----	-----
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistics	RX	TX	Total
-----	-----	-----	-----
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

Showing information when the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is down
Admin state is up
State information: Disabled by VSX
```

Link state: down for 3 days (since Tue Mar 16 05:20:47 UTC 2021)

Link transitions: 0

Description:

Hardware: Ethernet, MAC Address: 04:09:73:62:90:e7

MTU 1500

Type SFP+DAC3

Full-duplex

qos trust none

Speed 0 Mb/s

Auto-negotiation is off

Flow-control: off

Error-control: off

VLAN Mode: native-untagged

Native VLAN: 1

Allowed VLAN List: 1502-1505

Rate collection interval: 300 seconds

Rate	RX	TX	Total
(RX+TX)			

-----	-----	-----	
Mbits / sec	0.00	0.00	
0.00			
KPkts / sec	0.00	0.00	
0.00			
Unicast	0.00	0.00	
0.00			
Multicast	0.00	0.00	
0.00			
Broadcast	0.00	0.00	
0.00			
Utilization	0.00	0.00	
0.00			

Statistic	RX	TX
-----------	----	----

Total		
-----	-----	-----
Packets	0	
0	0	
Unicast	0	
0	0	
Multicast	0	
0	0	
Broadcast	0	

0	0	
Bytes		0
0	0	
Jumbos		0
0	0	
Dropped		0
0	0	
Pause Frames		0
0	0	
Errors		0
0	0	
CRC/FCS		0
n/a	0	
Collision		n/a
0	0	
Runts		0
n/a	0	
Giants		0
n/a	0	

Showing information when the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if) # show interface 1/1/1 physical
```


Control	EEE	Link	Admin	Speed		Flow-
Port	Type	PoE Power	Config	Status	Port	
Config	Status Config	Status (Watts)	State	Information	Config Description	Status

1/1/1	1GbT	up	up	1G	auto	off
off	on	on	--	10M/100M/1G	--	

Showing the monitor information:



NOTE
In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

```
Interface 1/1/1 is up
```

Rate	RX	TX	Total
(RX+TX)			

MBits / sec	30196.43	30196.43	
60392.85			
MPkts / sec	58977.39	58977.40	
117954.79			
Unicast	0.00	0.00	
0.00			
Multicast	58977.39	58977.40	
117954.79			
Broadcast	0.00	0.00	
0.00			
Utilization %	75.49	75.49	
150.98			
Statistic	RX	TX	Total
(RX+TX)			

Packets	4756527649	4756527865	
9513055514			
Unicast	0		
0	0		
Multicast	4756527649	4756527865	
9513055514			
Broadcast	2		
0	2		
Bytes	304417778668	304417795428	
608835574096			
Jumbos	0		
0	0		
Dropped	0	19028847730	
19028847730			
Pause Frames	0		
0	0		
Errors	0		
0	0		
CRC/FCS	0		
n/a	0		
help: ?, quit: q			
Help for Interface Monitor			
h Toggle human-readable mode			
c Clear interface statistics			
Does not apply to rates			
Arrows, PgUp, PgDn, Home, End			
Navigate interface statistics			

```
Delay: 2
help: ?, quit: q
```

Showing the output for interface 1/1/1 in human-readable format:



NOTE

In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```
switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up
Rate                                     RX                                     TX                                     Total
(RX+TX)
-----
Bits / sec                               3M
3M                                     6M
Pkts / sec                               316                                    316
633
Unicast                                 319                                    319
638
Multicast                               0
0                                     0
Broadcast                               0
0                                     0
Utilization %                           < 1                                    < 1
< 1
Statistic                               RX                                     TX
Total
-----
Packets                                 577K
577K                                   1M
Unicast                                 577K
577K                                   1M
Multicast                               0
51                                    51
Broadcast                               0
15                                    15
Bytes                                   744M
745M                                   1G
Jumbos                                 0
```

0	0	
Dropped		0
0	0	
Filtered		0
0	0	
Pause Frames		0
0	0	
Errors		0
0	0	
CRC/FCS		0
n/a	0	
Collision		n/a
0	0	
Runts		0
n/a	0	
Giants		0
n/a	0	

Showing information about extended counters:



NOTE

The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
```

```
-----
Interface 1/1/17
-----
```

Statistics	Value
Dot1d Tp Port In Frames	547
Dot1d Tp Port Out Frames	608
Dot3 In Pause Frames	0
Dot3 Out Pause Frames	0
Ethernet Stats Broadcast Packets	19
Ethernet Stats Bytes	40162
Ethernet Stats Packets	342

```
...
```

Error-Statistics	Value
Dot1d Base Port MTU Exceeded Discards	0
Dot3 Control In Unknown Opcodes	0
Dot3 Stats Alignment Errors	0

```

Dot3 Stats FCS Errors          0
Dot3 Stats Frame Too Longs     0
Dot3 Stats Internal Mac Transmit Errors 0
Ethernet RX Oversize Packets   0
...

```

Showing interface link-status:

```
switch# show interface link-status
```

```

-----
Port                Type                Physical      Link          Last
                  Link State    Transitions  Change
-----
1/1/1               1G-BT          down          0             --
1/1/2               1G-BT          up            1             1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/3               1G-BT          up            1             1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/4               --             down          0             --
1/1/5               --             down          0             --

```

Showing interface loopback 1 link-status:

```

-----
Port                Type                Physical      Link          Last
                  Link State    Transitions  Change
-----
loopback1           --             up            --            --

```

Showing interface 1/1/2-1/1/3 link-status:

```

-----
Port                Type                Physical      Link          Last
                  Link State    Transitions  Change
-----
1/1/2               1G-BT          up            1             1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)
1/1/3               1G-BT          up            1             1 minute ago (Fri
Mar 09 12:36:56 UTC 2018)

```

Showing interface link-status:

```
switch# show interface link-status
```

```

-----
Port                Type                Physical      Link          Link Flaps    Last
                  Link State    Transitions  Ignored      Change
-----
1/1/1               1G-BT          down          0             0             --
1/1/2               1G-BT          up            1             0             1 minute

```



```
ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3          1G-BT          up          1          0          1 minute
ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/4          --            down        0          0          --
1/1/5          --            down        0          0          --
```

Showing state information when interface is blocked:

```
8360(config-if)# show interface 1/1/1
Interface 1/1/1 is up (Blocked)
Admin state is up
State information: Blocked by UDLD
Link state: up for 1 minute (since Mon Jun 10 09:25:27 UTC 2024)
Link transitions: 1
Description:
Persona:
Hardware: Ethernet, MAC Address: 00:fd:45:67:85:91
MTU 1500
Type 10G-LR / 10G SFP+ LR
Full-duplex
qos trust none
Speed 10000 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rate collection interval: 300 seconds
```

Rate	RX	TX	Total
(RX+TX)			

Mbits / sec	0.00	0.00	
0.00			
KPkts / sec	0.00	0.00	
0.00			
Unicast	0.00	0.00	
0.00			
Multicast	0.00	0.00	
0.00			
Broadcast	0.00	0.00	
0.00			
Utilization %	0.00	0.00	
0.00			
Statistic	RX	TX	

Total			

Packets		15	
15	30		
Unicast		12	
12	24		
Multicast		3	
3	6		
Broadcast		0	
0	0		
Bytes		1350	1350
2700			
Jumbos		0	
0	0		
Dropped		0	
0	0		
Pause Frames		0	
0	0		
Errors		0	
0	0		
CRC/FCS		0	
n/a	0		
Collision		n/a	
0	0		
Runts		0	
n/a	0		
Giants		0	
n/a	0		

Command History

Release	Modification
10.15	Added state information when port goes into down state.
10.11	Added <code>monitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show interface downshift-enable

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] downshift-enable
```

Description

Displays speed downshift information, including the interface speed status and configuration.

Parameter	Description
<i><IFNAME></i>	Specifies a interface name.
<i><IFRANGE></i>	Specifies the port identifier range.

Examples

Showing automatic downshift information:

```
switch(config-if) # show interface downshift-enable
-----
Port          Downshift              Speed
             Enabled | Active      Status   | Config
-----
1/1/1         yes      yes      100M-FDx  auto
1/1/2         yes      no       1G        auto
```

1/1/3	yes	no	100M-FDx	100M-FDx
1/1/4	no	no	--	auto

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
```

Port	Downshift		Speed	
	Enabled	Active	Status	Config
1/1/2	yes	no	1G	auto

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
6000		
6100		

show running-config interface

Syntax

```
show running-config interface [<IFNNAME>|<IFRANGE>]
show running-config interface [lag | loopback | tunnel | vlan ] [<ID>]
```

Description

Displays active configurations of various switch interfaces.

Parameter	Description
<code><IFNAME></code>	Specifies a interface name.
<code><IFRANGE></code>	Specifies the port identifier range.
LAG	Specifies LAG interface information
LOOPBACK	Specifies loopback interface information.
TUNNEL	Specifies tunnel interface information.
VLAN	Specifies VLAN interface information.
<code><LAG-ID></code>	Specifies the LAG number. Range: 1 - 256.
<code><LOOPBACK-ID></code>	Specifies the LOOPBACK number. Range: 0 - 255.
<code><TUNNEL-ID></code>	Specifies the tunnel ID. Range: 1 - 255.
<code><VLAN-ID></code>	Specifies the VLAN ID. Range: 1 - 4094.
VXLAN	Specifies the VXLAN interface information.
<code><VXLAN-ID></code>	Specifies the VXLAN interface identifier. Default: 1.

Examples

Showing 1/1/2 interface configuration:

```
switch(config-if) # show running-config interface 1/1/2
interface 1/1/2
  no shutdown
  description DC-23
  exit
```

Showing loopback interfaces configured:

```
switch(config-if) # show running-config interface loopback
interface loopback 1
  description lb interface 1
  exit
interface loopback 2
  description lb interface 2
  exit
```

Showing loopback interfaces not configured:

```
switch(config-if) # show running-config interface loopback
No loopback interfaces configured.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Mirroring

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.



CAUTION

Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

This version of AOS-CX support the following mirror capabilities:

- Support for a VLAN as source for a mirror session
- 6100, 6000 and 4100i Switch series support of 2 simultaneously enabled Mirroring Sessions with LAG sources, regardless of direction
- Ability for a given Mirror source in one session to act as source in another Mirror Session
- Support for a Layer 2/bridged Link Aggregation Group (LAG) as Session destination
- Support for a Layer 3/Route-only Link Aggregation Group (LAG) as Session destination



NOTE

The following interface types are not supported as mirror source or destination:

- Management
- Persona
- Loopback
- VxLAN
- Subinterfaces

Hit Count Behavior for Policy Mirroring Actions

When using classifier policies with mirroring actions, it is important to note that packets mirrored via these policy actions will have their hit counts recorded within the Policy hit counts. These packets will not be recorded as output packets in the Mirror Session statistics, except in the case of Mirror-to-CPU sessions. This distinction means that traffic metrics for packets mirrored via Policy Actions should be obtained from Policy hit counts, as these packets will not appear in the Mirror Session output statistics.

Mirroring and CoPP

When configuring a Mirror Session with the destination set to the CPU, it is important to understand how mirrored traffic interacts with Control Plane Policing (CoPP).

Control plane packets that are already destined for the switch (for example, routing protocol packets and management traffic) will be processed by their respective CoPP classes. These packets will still be mirrored to the CPU as part of the Mirror-to-CPU session, but they will not be counted under the Mirror-to-CPU CoPP class. The Mirror-to-CPU CoPP class is specifically designed to manage dataplane traffic that is mirrored to the CPU. Any control plane traffic mirrored to the CPU will bypass this class and be handled by the CoPP class corresponding to its original purpose (e.g., OSPF, BGP, or ARP). Administrators should account for this distinction when analyzing traffic mirrored to the CPU.

For example, if a mirror session is configured to mirror all traffic from a source interface to the CPU, and that interface receives OSPF packets, the OSPF packets will be mirrored to the CPU. These packets will be processed by the OSPF CoPP class, not the Mirror-to-CPU CoPP class.

Subtopics

Mirror statistics

Classifier policies and mirroring sessions

Mirroring commands

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Classifier policies and mirroring sessions

About this task

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

Procedure

1. Configuring a mirroring session with a destination interface.
2. Enabling the mirroring session.
3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

Results

If the packets being mirrored are received from a VLAN that is not allowed on the mirror destination, the mirrored packets would be dropped at the mirror destination interface. When the mirrored packets are dropped at the destination, the mirror output packet and byte count will increment, however the packets will not be received at the mirror destination.

The mirror destination port among the active mirror sessions must be unique. That is, if an interface is configured as a source or destination in an active mirror session, the same port cannot be used as a destination in another active mirror session.

Scenario 1

Scenario 2

Mirroring commands

Subtopics

[clear mirror](#)

[comment](#)

[destination interface](#)

[destination tunnel](#)

[diagnostic](#)

[disable](#)

[enable](#)

[mirror endpoint](#)

[mirror session](#)

[show mirror](#)

[show mirror endpoint](#)

[source interface](#)

clear mirror

Syntax

```
clear mirror [all | <SESSION-ID>]
```

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Parameter	Description
all	Specifies all configured sessions.
<SESSION-ID>	Specifies a numeric identifier for the session. Range: 1 to 4

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

comment

Syntax

```
comment <COMMENT>
no comment
```

Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The **no** form of this command removes the comment.

Parameter	Description
<code><COMMENT></code>	A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3)# comment This Mirror will be removed during next maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3)# no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor statistics on each endpoint interfaces
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

destination interface

Syntax

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}  
no destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

Description

Configures the specified interface as the destination of the mirrored traffic.

The **no** form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Parameter	Description
<code><INTERFACE-ID></code>	Specifies a interface. Format: <code>member/slot/port</code> .
<code><LAG-NAME></code>	Specifies a LAG (link aggregation group) identifier.

Usage

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration. Mirroring traffic to a LAG destination with a device running LACP attached to the destination will cause the LAG link to flap if LACP packets are part of the mirrored traffic source.

Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1  
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

Switch	Destination interface limit per mirror session (4 possible sessions)
4100i	1
6000	1
6100	1
10040	1

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority

destination tunnel

Syntax

```
destination tunnel <TUNNEL-IPV4-ADDR> source <SOURCE-IPV4-ADDR>
    dscp <DSCP-VALUE> vrf <VRF-NAME> id <SPAN-ID>
no destination tunnel
```

Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror_err_tunnel_oob_port_not_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel_route_resolution_not_populated" operation status.



NOTE

The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

The **no** form of this command will cease the use of the tunnel and disable the session.

Parameter	Description
<code><TUNNEL-IPV4-ADDR></code>	Specifies the tunnel address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
	Specifies the source address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.

Parameter	Description
<code><SOURCE-IPv4-ADDR></code>	
<code><DSCP-VALUE></code>	Specifies the DSCP value to be carried within the DS field of E RSPAN packet header. Range: 0 to 63. Default: 0.
<code><VRF-NAME></code>	Specifies a VRF name. Default: default.

Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10
vrf default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp
10 vrf default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp
2 vrf default
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

diagnostic

Syntax

```
diagnostic  
diag utilities tshark [file]  
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.



NOTE

The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the **delete-file** form of this command to delete the most recent capture file.

Since **file** and **delete-file** are optional, the behavior of the base command **diag utilities tshark** does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Parameter	Description
file	Saves captured packets to a temporary file.
delete-file	Deletes the most recent captured file.

Example

Performing diagnostic:


```
switch# diagnostic
switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

disable

Syntax

```
disable
```

Description

Disables the mirroring session specified by the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the **show mirror** command for that session ID shows an **Admin Status** of **disable** and an **Operation Status** of **disabled**.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

enable

Syntax

```
enable
```

Description

Enables the mirroring session for the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the **show mirror** command for that session ID shows an **Admin Status** of **enable** and an **Operation Status** of **enabled**.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of

rx
or

both
):

- The attempt to enable the mirroring session fails and an error is returned.



NOTE

When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-mirror- <SESSION-ID>	Administrators or local user group members with execution rights for this command.

mirror endpoint

Syntax

```
mirror endpoint <NAME>
no mirror endpoint <NAME>
```

Description

Creates the specified mirror endpoint or enters its context if it already exists. The specifics of a mirror endpoint are created or altered while in the mirror endpoint context and the mirror endpoint is enabled or disabled from this context. It may be possible to support different encapsulations by different ASICs. For example, UDP for PVOS compatibility. Termination of GRE encapsulation is also supported.

The **no** form of this command removes an existing mirror endpoint. An enabled mirror endpoint is automatically disabled first before removal.

Parameter	Description
<NAME>	Specifies mirror endpoint name.

Examples

Creating a mirror endpoint named test :

```
switch(config) # mirror endpoint test
```

Deleting mirror endpoint named test:

```
switch(config) # no mirror endpoint test
```

Configuring a mirror endpoint named test :

```
6100(config) # mirror endpoint test
6100(config-mirror-endpoint-test) #
6100(config-mirror-endpoint-test) # destination
    interface Specify interfaces to send traffic
6100(config-mirror-endpoint-test) # destination interface
    IFNAMELIST An interface, a range or a comma seperated list of interfaces
6100(config-mirror-endpoint-test) # destination interface 1/1/3
    <cr>
6100(config-mirror-endpoint-test) # destination interface 1/1/3
6100(config-mirror-endpoint-test) #
6100(config-mirror-endpoint-test) # source 1.1.1.1 destination 1.1.1.2 id 1
```

```
vrf default
6100 (config-mirror-endpoint-test) #
```

**NOTE**

Only physical ports can be configured as interface for mirror-endpoint destination. LAG port is not supported as interface for mirror-endpoint destination.

**NOTE**

The maximum allowed number of destination interfaces for both mirror-session and mirror-endpoint is 1.

Command History

Release	Modification
10.13.1000	Added support for 4100i, 6000, and 6100 switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

mirror session

Syntax

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The **no** form of this command removes an existing mirroring session from the configuration.

Parameter	Description
<SESSION-ID>	Specifies the session identifier. Range: 1 to 4

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#
switch(config)# mirror session 3
switch(config-mirror-3)#
switch(config)# no mirror session 1
switch(config)#
```



NOTE

When configuring mirroring via the command-line interface, not all configuration errors will result in an immediate error message. After making configuration changes, always check the operation status of your mirror sessions using the **show mirror** command.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show mirror

Syntax

```
show mirror [<SESSION-ID>]
```

Description

Shows information about mirroring sessions. If `<SESSION-ID>` is not specified, then the command shows a summary of all configured mirroring sessions. If `<SESSION-ID>` is specified, then the command shows detailed information about the specified mirroring session.

Parameter	Description
<code><SESSION-ID></code>	Specifies the session identifier. Range: 1 to 4

Usage

Information in the **Admin Status** column of the command output indicates the configured status. The admin status is one of the following values:

- **enable**: The mirroring session is enabled.
- **disable**: The mirroring session has been configured but not yet enabled, or has been disabled.

Information in the **Operation Status** column indicates the status of the mirroring session. Operation status is one of the following values:

- **dest_doesnt_exist**: The configured destination interface is not found in the system. The mirroring session cannot be enabled.
- **destination_shutdown**: **The mirroring session is enabled, but the destination interface is shut down. No traffic** can be monitored.
- **disabled**: The mirroring session is disabled and is not in an error condition.
- **enabled**: The mirroring session is enabled.
- **external/driver_error**: An internal ASIC hardware error occurred.
- **hit_active_sessions_capacity**: The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.

- **internal_error:** An invalid parameter was passed to the ASIC software layer.
- **no_dest_configured:** The mirroring session does not have a destination interface configured.
- **no_name_configured:** A software error occurred. The mirroring session does not have a session ID in its configuration.
- **null_mirror:** A software error occurred. The session object reference is invalid.
- **out_of_memory:** The system is out of memory, reboot recommended.
- **tunnel_route_resolution_not_populated:** If the destination tunnel IP address is not reachable.
- **unknown_error:** An unexpected error occurred.

Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
---  -
1   enable       enabled
2   disable      disabled
3   disable      disabled
4   enable       internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
Source: interface 1/1/2 rx
Destination: interface 1/1/3
switch#
```

Show the details of mirror session 1 with an empty LAG as destination:

```
switch: show mirror 1
Admin Status: enable
Operation Status: dest_doesnt_exist
Source: interface 1/1/1 rx
Source: interface tx none
Destination: interface lag1
Output Packets: 0
Output Bytes: 0
```


Show the details of mirror session 1 with a LAG and a LAG member of same LAG as sources. In this scenario, traffic received on all LAG member interfaces is mirrored, is it is not necessary to define 1/1/1 as a source since it is already part of LAG1; this does not impact the mirror output packet or byte counters.

```
switch: show mirror 1
Admin Status: enable
Operation Status: enabled
Source: interface 1/1/1 rx
Source: interface lag1 rx
Destination: interface 1/1/2
Output Packets: 0
Output Bytes: 0
```

Showing the details of mirror **session 1** for a mirror configuration enabled with invalid interface which is not part of a lag:

```
switch# show mirror 1
Mirror Session: 1
Admin Status: enable
Operation Status: disabled
Source: interface 1/1/1 (unknown lag) both
Destination: cpu
Output Packets: 0
Output Bytes: 0
```

Certain configurations can cause this condition while the mirror source configuration remains unchanged:

- 1. Removing of a port from a LAG (Link Aggregation Group).
- 2. Port **split** or **unsplit** operations that result in LAG membership removal.

To recover from this issue, unconfigure the affected interface (previously part of a LAG), or restore the LAG membership configuration.

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mirror endpoint

Syntax

```
show mirror endpoint [<NAME>]
```

Description

Shows a list of all configured mirror endpoints, their Admin Status and their Operation Status.

The optional parameter will display the details of the specified mirror endpoint if it exists.

Parameter	Description
<NAME>	Specifies name of the mirror endpoint instance to be displayed.

Examples

Showing a summary of all configured mirror endpoints on the switch:

```
switch# show mirror endpoint
Name      Admin Status   Operation Status
-----  -
test      enable          enabled
monitor   disable         disabled
```

Showing the details of enabled mirror endpoint test:

```
switch# show mirror endpoint test
Mirror Endpoint: audit
Admin Status: enable
Operation Status: enabled
Comment: Mirror Endpoint Audit
Type: gre
Tunnel: source 1.1.1.1 destination 1.1.1.2 id 1 vrf default
Interface: 1/1/3
```

Output Packets: 123456789

Output Bytes: 0



NOTE

"Output Packets" in "show mirror endpoint [name]" is only supported for statistics.

"Output Bytes" in "show mirror endpoint [name]" is not supported due to ASIC limitation.

Command History

Release	Modification
10.13.1000	Added support for 4100i, 6000, and 6100 switches.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (<code>></code>) or Manager (<code>#</code>)	Administrators or local user group members with execution rights for this command.

source interface

Syntax

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored.

The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Parameter	Description
<code><PORT-NUM></code>	Specifies a physical port on the switch. Use the format member/slot/port (for example, 1/3/1).
<code><LAG-NAME></code>	Specifies the identifier for the LAG (link aggregation group).
<code><DIRECTION></code>	Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following: • both: Mirror both transmitted and received packets. • rx: Mirror only received packets. • tx: Mirror only transmitted packets.

Usage

There is a limit of source interfaces in each direction of a given mirror session:

Switch	Source interface limit per mirror session (4 possible sessions)
4100i	28
6000	52
6100	52
10040	128

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.



NOTE

When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface  
LAG-NAME      Enter a LAG name. For example, lag10  
PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1  
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2  
switch(config-mirror-2)# source interface 1/1/3 tx  
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror- <SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

Monitoring a device using SNMP

Configuring SNMP: Refer to the SNMP/MIB Guide for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the SNMP/MIB Guide and specific information about the `show snmp trap` command to enable SNMP traps.

Packet Capture

The Packet Capture is a solution which enables the Central administrator to capture or view packets from clients connected to Halon switches. The solution serves as a tool which an admin can leverage to debug clients connected to managed devices across branches in an enterprise. The switch requires packet exchanges to be copied to the CPU (both ingress and egress packets) per client session, so that they can be streamed to Central where the admin can view the packet exchanges for further analysis.

This feature is supported on the following platforms:

- Rosewood (6200, 6300, 6400, 8360)
- Matrix (4100i, 6000, 6100)

Power-over-Ethernet

- The Power-over-Ethernet (PoE) subsystem manages power supplied to devices using standard Ethernet data cables. A Power Sourcing Equipment (PSE) supplies DC power as well as Ethernet connectivity to a Powered Device (PD) using a standard Ethernet cable. The maximum current depends on the PD Requested Class.
- A PoE subsystem contains two parts : a PSE and PD. A Power Sourcing Equipment (PSE) is a device that provides power through a standard Ethernet cable. A PoE capable switch functions as PSE. All Aruba PoE switches are considered as PSEs. A PD is a device powered by a PSE. Examples of PD are VoIP phones, Wireless APs, and IP cameras.
- When a PD or any network cable is connected to a PSE port, the PSE applies a detection voltage and measures the resistance value of the PD. If resistance is within IEEE 802.3 standard values (23 - 26k ohm), the connected device is treated as PD and classification begins. For legacy devices to be detected, you must enable prestandard detection on the switch.
- PDs are divided into different types and classes based on PD power requirements. The power supplied by the PSE is higher than the power PD draws to accommodate for the line losses that can result with the use of the standard maximum length cable(100m).
 - Type 1: PSE can supply maximum of 15.4W, and PD can draw a maximum of 13W.
 - Type 2: PSE can supply maximum of 30W, and PD can draw a maximum of 25.5W.
 - Type 3: PSE can supply maximum of 60W, and PD can draw a maximum of 51W.
 - Type 4: PSE can supply maximum of 90W, and PD can draw a maximum of 71W.
- Classes of PD:
 - Class 0: Type1 PD, it can draw a maximum of 13W.
 - Class 1: Type1 PD, it can draw a maximum of 3.84W.
 - Class 2: Type1 PD, it can draw a maximum of 6.49W.
 - Class 3: Type1 PD, it can draw a maximum of 13W.
 - Class 4: Type2 PD, it can draw a maximum of 25.5W.
 - Class 5: Type3 PD, it can draw a maximum of 40W.
 - Class 6: Type3 PD, it can draw a maximum of 51W.
 - Class 7: Type4 PD, it can draw a maximum of 62W.
 - Class 8: Type4 PD, it can draw a maximum of 71.3W.

- IEEE 802.3bt introduced 4-Pair PoE as a means of supplying higher power to PDs that need more than the current 25.5W supplied by IEEE 802.3at. To increase the available power without damaging the Ethernet cable, the standard introduced the ability to use all four pairs within the Ethernet cable instead of the two pairs used by previous standards (802.3at, 802.3af).
- Supported protocols:
 - Compatibility with IEEE 802.3af, 802.3at, 802.3bt and prestandard.
 - Long first class event supported on Type 3-4 PSE.
 - Support for Single Signature (SS) Type 0-6 and Dual Signature (DS) Type 0-4 PDs.
 - Multi-Event classification permits mutual ID of SS Class 0-6 and DS Class 0-4.
 - Support LLDP Data Link Layer (DLL) Type 1-2 extension 12-octet TLV and Type 3-4 extension 29-octet TLV.
 - Default PSE assigned class delivers the maximum PSE capable power at initial power up based on PD requested class.
- Always-on PoE is a feature that provides the ability for a switch to continue to provide power across user initiated reboots through software. Always-on PoE is enabled by default and no additional configuration is needed.



NOTE

PDs only remain powered, no data transfer or PoE power negotiation can occur until the switch has completely booted up and in normal operation. PD faults occurring prior to full switch boot up will result in PoE power removal and restart the detection process only after switch returns to normal operation.

Subtopics

PoE commands

PoE commands

All PoE configuration commands except **quick-poe**, **threshold configuration** and **always-on poe configuration** are entered at the **config-if** context. The PoE threshold command is used at the system level whereas the **always-on poe** and **power-over-ethernet quick-poe** commands are set at the slot level. These commands can only be configured in the global configuration context.

Subtopics

lldp dot3 poe

lldp med poe
power-over-ethernet
power-over-ethernet allocate-by
power-over-ethernet always-on
power-over-ethernet assigned-class
power-over-ethernet pre-std-detect
power-over-ethernet priority
power-over-ethernet quick-poe
power-over-ethernet threshold
power-over-ethernet trap
show lldp local
show lldp neighbor
show power-over-ethernet

lldp dot3 poe

Syntax

```
lldp dot3 poe  
no lldp dot3 poe
```

Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The **no** form of this command disables 802.3 TLV list in LLDP.

Examples

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1  
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

lldp med poe

Syntax

```
lldp med poe [priority-override]  
no lldp med poe [priority-override]
```

Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The **no** form of this command disables MED TLV list in LLDP.

Parameter	Description
[priority-override]	System defined name of the interface.

Examples

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1  
switch(config-if)# lldp med poe
```

```
switch(config-if) # no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if) # lldp med poe priority-override
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet

Syntax

```
power-over-ethernet  
no power-over-ethernet
```

Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module.

The **no** form of this command disables per-interface power distribution.

Examples

Enabling per-interface power distribution:

```
switch(config) # interface 1/1/1  
switch(config-if) # power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if) # no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config) # power-over-ethernet quick-poe 1/1
switch(config) # no power-over-ethernet
Interface PoE cannot be disabled when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet allocate-by

Syntax

```
power-over-ethernet allocate-by {usage | class}
no power-over-ethernet allocate-by {usage | class}
```

Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The power allocation method can be changed on an interface through port-access (User roles or RADIUS). An allocation method when configured through port-access will replace the user configured method.

The **no** form of this command resets the action to default.

Parameter	Description
usage	Configures the usage - based allocation method.
class	Configures the class - based allocation method.

Usage

If you enable **pd-class-override** for an interface, the **allocate-by** configuration of that interface will be automatically changed to **class**. However, if you change the allocation method to **usage** when **pd-class-override** is still enabled, you will receive an error message stating that "The power allocation method cannot be changed when pd-class-override is enabled."

To remove **pd-class-override**, you can use the **no power-over-ethernet pd-class-override** command . It is important to note that **pd-class-override** requires the allocation method to be set to **class** and is enforced when configured through CLI. However, if you override the allocation method to **usage** via port-access, **pd-class-override** will not be in effect. Therefore, it is recommended that you do not override the allocation method to **usage** through port-access on interfaces configured with **pd-class-override**.

Examples

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

Command History

Release	Modification
10.07 or earlier	- -

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

power-over-ethernet assigned-class

Syntax

```
power-over-ethernet assigned-class {3 | 4 | 6}  
no power-over-ethernet assigned-class
```

Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the subsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The **no** form of this command resets the action to default.

Examples

Setting PoE assigned class:

```
switch(config)# interface 1/1/1  
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config) # power-over-ethernet quick-poe 1/1
switch(config) # interface 1/1/1
switch(config) # power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet pre-std-detect

Syntax

```
power-over-ethernet pre-std-detect
no power-over-ethernet pre-std-detect
```

Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. HPE Aruba Networking switches are backward compatible and will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The **no** form of this command resets the action to default.

Examples

Enabling pre-standard device detection:


```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling pre-standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet priority

Syntax

```
power-over-ethernet priority {critical | high | low}
no power-over-ethernet priority {critical | high | low}
```

Description

Sets PoE priority for an interface. Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet quick-poe

Syntax

```
power-over-ethernet quick-poe <MODULE-ID>
no power-over-ethernet quick-poe <MODULE-ID>
```



NOTE

This command is not available on the 6000 or 6100 Switch Series.



NOTE

This command is available only on the 4100i Switch Series.

Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The **no** form of this command disables quick PoE.

Parameter	Description
<code><MODULE-ID></code>	Specifies module number for quick PoE configuration .

Examples

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/2
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet threshold

Syntax

```
power-over-ethernet threshold <PERCENTAGE>
no power-over-ethernet threshold <PERCENTAGE>
```

Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The **no** form of this command resets the action to default.

Parameter	Description
<PERCENTAGE>	Excess power consumption trap threshold. Range 1 - 99.

Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000	config	Administrators or local user group members with execution rights for this command.

Platforms	Command context	Authority
6100		

power-over-ethernet trap

Syntax

```
power-over-ethernet trap
no power-over-ethernet trap
```

Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Enabling SNMP trap generation for PoE:

```
switch(config) # power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if) # no power-over-ethernet trap
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i	config-if	Administrators or local user group members with execution rights for this command.
6000		
6100		

show lldp local

Syntax

```
show lldp local-device [<INTERFACE-ID>]
```

Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

Parameter	Description
<code><INTERFACE-ID></code>	Specifies an interface. Format: member/slot/port

Examples

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
=====
Port-ID           : 1/1/10
Port-Desc         : "1/1/10"
Port VLAN ID      : 0
PoE Plus Information
PoE Device Type   : Type 2 PSE
Power Source       : Primary
Power Priority     : low
PSE Allocated Power : 25.0 W
PD Requested Power : 25.0 W
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show lldp neighbor

Syntax

```
show lldp neighbor [<INTERFACE-ID>]
```

Description

Displays detailed information about a particular neighbor connected to a particular interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port

Examples

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10
Port                                     : 1/1/10
Neighbor Entries                        : 1
Neighbor Entries Deleted                : 0
Neighbor Entries Dropped                : 0
Neighbor Entries Aged-Out               : 0
Neighbor Chassis-Name                   : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description            : ArubaOS (MODEL: 325), Version HPE_ANW IAP
Neighbor Chassis-ID                     : 84:d4:7e:ce:5d:68
Neighbor Management-Address             : 169.254.41.250
Chassis Capabilities Available          : Bridge, WLAN
Chassis Capabilities Enabled            :
```

```

Neighbor Port-ID           : 84:d4:7e:ce:5d:68
Neighbor Port-Desc         : eth0
TTL                         : 120
Neighbor Port VLAN ID      :
Neighbor PoEplus information : DOT3
Neighbor Device Type       : TYPE2 PD
Neighbor Power Priority     : Unkown
Neighbor Power Source      : Primary
Neighbor Power Requested   : 25.0 W
Neighbor Power Allocated   : 0.0 W
Neighbor Power Supported   : No
Neighbor Power Enabled     : No
Neighbor Power Class       : 5
Neighbor Power Paircontrol : No
Neighbor Power Pairs       : SIGNAL

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show power-over-ethernet

Syntax

```
show power-over-ethernet [member <MEMBER-ID>] [brief]
```

Description

Displays the status information of the full system.

Parameter	Description
<code><MEMBER-ID></code>	Displays the detailed status of given member.
<code>brief</code>	Display the brief status of all ports or the given port.

Examples

Showing sample output for show power-over-ethernet :

```
switch# show power-over-ethernet
System Power Status
  PoE Power Status           : No redundancy
  Operational Power Status    : No redundancy
  Total Available Power       : 360 W
  Total Configured Power      : 0 W
  Total Failover Pwr Avl      : 0 W
  Total Redundancy Power      : 0 W
  Total Power Drawn           : 0 W
  Total Power Reserved        : 0 W
  Total Remaining Power       : 360 W
  Trap Threshold              : 80 %
  Trap Enabled                : Yes
  Always-on PoE Enabled       : 1/1
  Quick PoE Enabled           : None
Internal Power
  Total Power
  PS      (Watts)      Status
  ----  -
  1/1    0             Ok
  1/2    0             Absent
```

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief
Status and Configuration Information for port 1/1/1
Member 1Power Status
  Available: 370 W  Reserved: 55.60 W  Remaining: 314.40 W
  Always-on PoE Enabled: 1/1
PoE      Pwr Power      Pre-std Alloc PSE Pwr PD Pwr PoE Port      PD      Cls
Type
Port     En  Priority Detect  Act   Rsrvd  Draw  Status      Sign
```

-----	---	-----	-----	-----	-----	-----	-----	-----	---	---
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2

Showing sample output for power-over-ethernet brief for interface range:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief
Status and Configuration Information for port 1/1/1-1/1/2
  Power Status
  Available: 360 W   Reserved: 0.00 W   Remaining: 360.00 W
  Always-on PoE Enabled: 1/1
  Quick PoE Enabled: None
PoE      Pwr Power      Pre-std Alloc PSE Pwr PD Pwr PoE Port      PD      Cls
Type
Port      En  Priority Detect   Act    Rsrvd   Draw   Status      Sign
-----
1/1/1     Yes Low          Off      Usage  0.0 W   0.0 W Searching  N/A      N/A
N/A
1/1/2     Yes Low          Off      Usage  0.6 W   0.0 W Searching  N/A      N/A
N/A
```

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief
Module 1/3 is not physically present.
```

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1
Interface 2/1/1 is not present.
```

Command History

Release Modification

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
4100i 6000 6100	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Aruba AirWave

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the AOS-CX Release Notes.

Subtopics

SNMP support and AirWave

Supported features with AirWave and the AOS-CX switch

Configuring the AOS-CX switch to be monitored by AirWave

AirWave commands

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config`

command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
snmp-server vrf mgmt
!
...
```

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface.
	Firmware versions
	Displays neighbor devices connected to AOS-CX switches
	Device topology

Configuration management	Partial configuration
Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the switch by entering the `snmp-server vrf` command.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBApZHf2d20GYr/xCGUzYzm0zjNf/4VKHtSqBNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBApb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
switch(config)# snmpv3 context Admin
```

For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as mgmt. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as default or mgmt.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

AirWave commands

Subtopics

[Remote syslog commands](#)

[SNMP commands](#)

Remote syslog commands

Subtopics

[clear accounting-logs](#)

[diag persistent-storage](#)

[logging](#)

[logging accounting-format-native](#)

[logging filter](#)

[logging facility](#)

[logging persistent-storage](#)

```
clear accounting-logs
```

Syntax

```
clear accounting-logs
```

Description

Use this command to clear accounting logs. Once issued, only logs generated after this command is run will be displayed in the output of the **show accounting log** commands.



NOTE

This command will not clear logs when the logging accounting-format-native feature is configured. To clear accounting logs on switches with this feature enabled, users should first revert the native accounting format back to the default AOS-CX format by executing the **no logging accounting-format-native** command.

Example

```
switch(config)# clear accounting-logs
```

The following example shows that accounting logs cannot be cleared using the clear accounting-logs command if the **logging accounting-native-format** command has been enabled, and that disabling this option with the **no logging accounting-format-native** command again allows the accounting logs to be cleared.

```
switch# logging audit-format-native
switch# clear accounting-logs
Warning: Clear accounting-logs is not supported for 'audit-format-native'.
switch# no logging audit-format-native
switch# clear accounting-logs
switch# show accounting log last 5
-----
Command logs from current boot
-----
No command logs has been logged in the system
```

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

diag persistent-storage

Syntax

```
diag persistent-storage enable [{severity <LEVEL> | timeout <TIMEOUT> |  
filter <FILTER_NAME> | storage-writes <STORAGE_WRITES> | usb <STORAGE_URL>}]  
diag persistent-storage disable
```

Description

Enables or disables storage of logs in storage. Only logs of the specified severity and above will be preserved in the storage.



NOTE

To switch the persistent storage setting from one option (e.g., USB) to another (e.g., storage-writes), you must first disable persistent storage by running the command `diag persistent-storage disable`. That the storage-writes option applies only to local storage and does not work with USB.

Parameter	Description
enable	Enable persistent logging.
disable	Disable persistent logging.
severity <LEVEL>	Specifies the severity of the syslog messages: • alert: Preserves syslog messages with the severity of alert (6) and emergency (7) • crit: Preserves syslog messages with the severity of critical (5) and above. Default. • debug: Preserves syslog messages with the severity of debug (0) and above.

Parameter	Description
	• emer: Preserves syslog messages with the severity of emergency (7) only. • err: Preserves syslog messages with the severity of err (4) and above. • info: Preserves syslog messages with the severity of info (1) and above. • notice: Preserves syslog messages with the severity of notice (2) and above. • warn: Preserves syslog messages with the severity of warning (3) and above.
<code>filter <FILTER_NAME></code>	Preserve logs for specified filter name.
<code>timeout <TIMEOUT></code>	Preserve logs for specified duration. Range: 1 to 60 minutes. Default: 20 minutes.
<code>storage-writes <STORAGE_WRITES></code>	Limit the amount of storage writes.
<code>usb <STORAGE_URL></code>	Preserve logs in the USB storage.

Examples

Enabling storage of logs in storage with severity **err** and timeout 30 minutes:

```
switch(config)#diag persistent-storage enable severity err timeout 30
Logs will be written to storage. Prolonged usage of persistent logging can
affect lifetime of storage. Hence enable persistent logging only for short
duration and also for the right severity.
Do you want to continue (y/n)? y
```

Enabling storage of logs in storage with severity **err** and **timeout** 30 minutes and 512 storage-writes:

```
switch(config)#diag persistent-storage enable severity err timeout 30
storage-writes 512
Logs will be written to storage. Prolonged usage of persistent logging can
affect lifetime of storage. Hence enable persistent logging only for short
duration and also for the right severity.
Do you want to continue (y/n)? y
```

Disabling log storage:

```
switch(config)# diag persistent-storage disable
```

Command History

Release	Modification
10.17	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

logging

Syntax

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME>} {udp [<PORT-NUM>]} |  
{tcp [<PORT-NUM>]} | {tls [<PORT-NUM>]}  
    auth-mode {certificate|subject-name}  
    disable  
    filter <FILTER-NAME>  
    include-auditable-events  
    legacy-tls-renegotiation]  
    rate-limit-burst <BURST>  
    rate-limit-interval <INTERVAL>] ]  
    severity <LEVEL>]  
    vrf <VRF-NAME>]  
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME> }
```

Description

Enables syslog forwarding to a remote syslog server.

The **no** form of this command disables syslog forwarding to a remote syslog server.

Starting with AOS-CX 10.11, payload information is present in accounting logs.

The maximum REST payload that can be sent to RADIUS/TACACS server is 1024 characters, and the maximum of REST payload that can be sent to syslog server is 3500 characters. If this limit is reached, the log will display three dots (...) to indicate that the log exceeded the character limit and is incomplete.

Parameter	Description
<code>{ <IPV4-ADDR> <IPV6-ADDR> <HOSTNAME> }</code>	Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.
<code>[udp [<PORT-NUM>] tcp [<PORT-NUM> tls [<PORT-NUM>]]</code>	Specifies the UDP port, TCP port, or TLS port of the remote syslog server to receive the forwarded syslog messages.
<code>udp [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 514
<code>tcp [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 1470
<code>tls [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 6514
<code>auth-mode</code>	Specifies the TLS authentication mode used to validate the certificate. • certificate: Validates the peer using trust anchor certificate based authentication. Default. • subject - name: Validates the peer using trust anchor certificates as well as subject - name based authentication.
<code>disable</code>	Disable remote syslog configuration. This does not delete the configuration, just disables/pauses the forwarding of syslog messages to the remote server. The config/forwarding can be reenabled (un - paused) again using the no logging <hostname> disable command.
<code>filter <FILTER-NAME></code>	Specifies the name of the filter to be applied on the syslog messages.
<code>include-auditable-events</code>	Specifies that auditable messages are also logged to the remote syslog server.
<code>legacy-tls-renegotiation</code>	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.
<code>rate-limit-burst <BURST></code>	Specifies the rate limit for the messages sent to the remote syslog server.
<code>rate-limit-interval <INTERVAL></code>	Specifies the rate limit interval in seconds. Default: 30 Seconds
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: • alert: Forwards syslog messages with the severity of alert (6) and emergency (7).

Parameter	Description
	• crit: Forwards syslog messages with the severity of critical (5) and above. • debug: Forwards syslog messages with the severity of debug (0) and above. • emerg: Forwards syslog messages with the severity of emergency (7) only. • err: Forwards syslog messages with the severity of err (4) and above. • info: Forwards syslog messages with the severity of info (1) and above. Default. • notice: Forwards syslog messages with the severity of notice (2) and above. • warning: Forwards syslog messages with the severity of warning (3) and above.
<code>vrf <VRF-NAME></code>	Specifies the VRF used to connect to the syslog server. Optional. Default: <code>default</code>

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of **err (4)** and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF **lab_vrf**:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject-name
```

Applying log filtering for syslog server forwarding:

```
switch(config)# logging 10.0.10.6 severity info filter filter_lldp_logs vrf mgmt
```

Applying log filtering and enabling the rate limit for syslog server forwarding over TCP port:

```
switch(config)# logging 10.0.10.2 tcp 3440 severity err vrf mgmt include-  
auditable-events filter filter_lldp_logs rate-limit-burst 3 rate-limit-  
interval 35
```

Command History

Release	Modification
10.12.1000	The disable parameter is introduced
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logging accounting-format-native

Syntax

```
logging accounting-format-native  
[no] logging accounting-format-native
```

Description

Change the accounting log message format to native Linux format. (Default: AOS-CX format)

The 'no' form of this command will change the accounting log message format to AOS-CX format.

Usage

This option enables the switch to show all types of accounting records to the user. When configured, the same format will be used while sending messages to syslog servers. When upgrading to this version of AOS-CX from AOS-CX 10.10 or earlier versions, if native accounting logs are preferred, then best practices is to issue this command as a part of the upgrade. If the switch upgrades from AOS-CX 10.10 or earlier without configuring this setting, by default, the accounting log message format will be AOS-CX Format.

Example

This example changes the accounting log message format to native Linux format.

```
switch(config)# logging accounting-format-native
```

The following example returns the accounting log message format to the default AOS-CX format.

```
switch(config)# no logging accounting-format-native
```

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logging filter

Syntax

```
logging filter <FILTER-NAME>
    [{enable | disable}]
    [<SEQUENCE-ID> {permit | deny} [event-id <EVENT-ID-RANGE>] [includes
<REGEX>] [severity <COMPARISON-OPERATOR>
    <LEVEL>]
    no <SEQUENCE-ID>
    resequence <OLD-SEQUENCE-ID>
    <NEW-SEQUENCE-ID>
no logging filter <FILTER-NAME>
```

Description

Creates a filter to restrict what event or debug logs are logged. A filter can be used to either permit or deny:

- The event logs from being generated on the switch, or
- The event or debug logs generated on the switch from being forwarded to a syslog server.

A filter is identified by a filter name and can have up to 20 rules or entries, each with a different sequence number, matching criteria, and corresponding action (deny or permit). When a filter is applied on a log, the log is matched against the criteria mentioned in the rules or entries in ascending numerical order of their sequence numbers until a matching entry is found. Once a matching entry is found, its corresponding action is applied on the log. If no matching rule is found, the default action (permit) is applied.

The **no** form of this command removes the filter.

Parameter	Description
<code><FILTER-NAME></code>	Specifies the unique name to identify the filter.
<code>enable</code>	Filter event logs generated on the switch.
<code><SEQUENCE-ID></code>	Specifies the filter criteria sequence number. Default: Increments by 10 from the largest sequence - id currently used in this filter.
<code>deny</code>	Prevents the matching log from being logged.
<code>permit</code>	Allows the matching log.
<code><event-id></code>	Matches logs by event ID. Specify an event ID or a range of event IDs. It supports a maximum of 100 event IDs.
<code>includes <REGEX></code>	Matches the log message against a regular expression string.
<code>severity</code>	Matches the logs by severity level.

The following options are used to compare the severity:

- **eq:** Match events of severity equal to the specified.
- **ge:** Match events of severity greater than or equal to the specified.
- **gt:** Match events of severity greater than the specified.
- **le:** Match events of severity lesser than or equal to the specified.
- **lt:** Match events of severity lesser than the specified.

Parameter	Description
	The following are the severity levels: • alert: Logs with the severity alert (6). • crit: Logs with the severity critical (5). • debug: Logs with the severity debug (0). • emerg: Logs with the severity emergency (7). • err: Logs with the severity err (4). • info: Logs with the severity info (1). • notice: Logs with the severity notice (2). • warning: Logs with the severity warning (3).

Usage

Filtering event logs on the switch: To permit or deny event logs from being generated on the switch. In this case, the matching event logs are filtered at generation. The denied event logs are neither logged to the switch events nor forwarded to any remote syslog servers. Multiple filters can be configured, but only one filter can be applied to filter the events on the switch. Such a filter can be chosen by adding the **enable** command under its configuration. Configuring the **enable** command under a new filter automatically removes it from the filter where it was previously used.

For example:

```
logging filter low_severity_logs
enable
10 deny severity lt info
```

This configuration denies the event logs which have a severity less than info.



NOTE

If a filter contains **enable** command, it is not recommended to configure this filter in the **logging** command used for remote syslog server configuration. This is because, any event logs denied by the filter are already not available for forwarding to a remote server.

A filter with **enable** command will not affect debug logs. Consider the configuration in the following example of a filter with **enable** command and two rules applied **10 permit severity ge info** and **20 deny**. This implies permit only those event logs which have severity greater than or equal to **info**.

Example:

```
logging filter low_severity_logs
enable
10 permit severity ge info
20 deny
```


Filtering event or debug logs when forwarding to a remote syslog server: The filter name must be configured in the logging command that is used to configure remote syslog server. The logs will be generated on the switch and the filter only decides whether to deny or permit the syslog forwarding for the matching log. For example: **logging 10.0.10.6 filter filter_ldp_logs**



NOTE

The filter affects debug logs only when the command **debug destination syslog** is configured on the switch.



NOTE

The severity mentioned in the remote syslog server configuration using logging command under configuration context has more precedence than the severity mentioned in a filter entry. If a log with warning severity is permitted by a filter, but the remote syslog configuration has severity **err** mentioned in it, the log will not be forwarded to the remote syslog server (since warning(3) is lesser than err(4)). On the other hand, if a log with **err** severity is permitted by a filter and the remote syslog configuration has severity warning mentioned in it, the log will be forwarded to the remote syslog server.

Examples

Configuring a new logging filter:

```
switch(config)# logging filter example_filter
```

To deny logs having event ID 1301 and a range of event IDs from 1305 to 1309:

```
switch(config-logging-filter)# 20 deny event-id 1301,1305-1309
```

To permit logs having event ID 1300:

```
switch(config-logging-filter)# 30 permit event-id 1300
```

To permit logs with severity greater than or equal to `err`:

```
switch(config-logging-filter)# 30 permit severity ge err
```

To deny logs with severity greater than info:

```
switch(config-logging-filter)# 30 deny severity gt info
```

To deny logs with event ID 1024 and a message matching the regular expression LLDP:

```
switch(config-logging-filter)# 40 deny event-id 1024 includes LLDP
```

Denying all logs:

```
switch(config-logging-filter)# 40 deny
```

Changing the sequence ID of an existing rule:

```
switch(config-logging-filter)# resequence 20 70
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config and config-logging-filter	Administrators or local user group members with execution rights for this command.

logging facility

Syntax

```
logging facility {local0 | local1 | local2 | local3 | local4 | local5 |  
local6 | local7}  
no logging facility
```

Description

Sets the logging facility to be used for remote syslog messages. Default: local7

The **no** form of this command disables the logging facility to be used for remote syslog messages.

Parameter	Description
<pre>{local0 local1 local2 local3 local4 local5 local6 local7}</pre>	Selects the logging facility to be used for remote syslog messages. Required. Specifies the severity of the syslog messages: • local0 • local1 • local2 • local3 • local4 • local5 • local6 • local7

Examples

Sets the local5 logging facility to be used for remote syslog messages:

```
switch(config)# logging facility local5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

logging persistent-storage

Syntax

```
logging persistent-storage [severity {alert|crit|debug|emerg|err|info|
notice|warning}]
no logging persistent-storage
```

Description

Enables or disables storage of logs in storage. Only logs of the specified severity and above will be preserved in the storage.

The **no** form of this command disables storage of logs in storage.



NOTE

This command is deprecated. Please use "diag persistent-storage" command to preserve logs in storage.

Parameter	Description
severity <LEVEL>	Specifies the severity of the syslog messages: • alert: Preserves syslog messages with the severity of alert (6) and emergency (7) • crit: Preserves syslog messages with the severity of critical (5) and above. Default. • debug: Preserves syslog messages with the severity of debug (0) and above. • emerg: Preserves syslog messages with the severity of emergency (7) only. • err: Preserves syslog messages with the severity of err (4) and above. • info: Preserves syslog messages with the severity of info (1) and above. • notice: Preserves syslog messages with the severity of notice (2) and above. • warning: Preserves syslog messages with the severity of warning (3) and above.

Usage

These logs can be copied out by using the **copy support-files all** or **copy support-files previous-boot**.

Examples

Enabling storage of logs in storage with severity **info**:

```
switch(config)#logging persistent-storage severity info  
Logs will be written to storage and made available across reboot.  
Do you want to continue (y/n)?
```

Disabling storage of logs in storage:

```
switch(config)# no logging persistent-storage
```

Command History

Release	Modification
10.17	Command deprecated.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

SNMP commands

Subtopics

- [event-trap-enable](#)
- [lldp trap enable](#)
- [mac-notify traps](#)
- [rmon alarm {enable | disable} {index | all}](#)
- [rmon alarm](#)
- [show configuration-changes trap](#)
- [show mac-notify port](#)
- [show mac-notify](#)
- [show rmon alarm](#)
- [show snmp agent-port](#)
- [show snmp community](#)
- [show snmpv3 context](#)
- [show snmpv3 engine-id](#)
- [show snmpv3 security-level](#)

show snmp system
show snmp trap
show snmpv3 users
show snmp views
show snmp vrf
snmpv3 context
snmpv3 engine-id
snmpv3 security-level
snmp-server agent-port
snmp-server community view
snmp-server community
snmp-server historical-counters-monitor
snmp-server host
snmp-server system-contact
snmp-server system-description
snmp-server system-location
snmp-server trap-source interface vrf
snmp-server vrf
snmpv3 user view
snmpv3 user
snmp-server response-source
snmp-server snmpv3-only
snmp-server trap
snmp-server trap aaa-server-reachability-status
snmp-server trap configuration-changes
snmp-server trap mac-notify
snmp-server trap port-security
snmp-server trap snmp
snmp-server trap vsx
snmp-server view

event-trap-enable

Syntax

```
event-trap-enable  
no event-trap-enable
```

Description

Enables the notification of events to be sent as traps to the SNMP management stations. It is enabled by default.

The **no** form of this command disables the event traps.

Examples

Enabling the event traps:

```
switch(config) # event-trap-enable
```

Disabling the event traps:

```
switch(config) # no event-trap-enable
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

lldp trap enable

Syntax

```
lldp trap enable  
no lldp trap enable
```

Description

Enables sending SNMP traps for LLDP related events from a particular interface. LLDP trap generation is enabled by default on all the interfaces and has to be disabled for interfaces on which traps are not required to be generated.

The **no** form of this command disables the LLDP trap generation.



NOTE

LLDP trap generation is disabled by default at the global level and must be enabled before any LLDP traps are sent.

Examples

Enabling LLDP trap generation on global level:

```
switch(config)# lldp trap enable
```

Enabling LLDP trap generation on interface level:

```
switch(config-if)# lldp trap enable
```

Disabling LLDP trap generation on global level:

```
switch(config)# no lldp trap enable
```

Disabling LLDP trap generation on interface level:

```
switch(config-if)# no lldp trap enable
```

Displaying LLDP global configuration:

```
switch# show lldp configuration
```

```
LLDP Global Configuration
```

```
=====
```

```
LLDP Enabled : No
```

```
LLDP Transmit Interval : 30
```

```
LLDP Hold Time Multiplier : 4
```

```
LLDP Transmit Delay Interval : 2
```

```
LLDP Reinit Timer Interval : 2
```

```
LLDP Trap Enabled : No
```

```
TLVs Advertised
```

```
=====
```

```
Management Address
```

```
Port Description
```

```
Port VLAN-ID
```

```
System Description
```

```
System Name
```

```
LLDP Port Configuration
```

```
=====
```

PORT	TX-ENABLED	RX-ENABLED	INTF-TRAP-ENABLED
------	------------	------------	-------------------

1/1/1	Yes	Yes	Yes
-------	-----	-----	-----

1/1/2	Yes	Yes	Yes
-------	-----	-----	-----

1/1/3	Yes	Yes	Yes
-------	-----	-----	-----

1/1/4	Yes	Yes	Yes
-------	-----	-----	-----

1/1/5	Yes	Yes	Yes
-------	-----	-----	-----

1/1/6	Yes	Yes	Yes
-------	-----	-----	-----

```
.....
```


.....			
mgmt	Yes	Yes	Yes

Displaying LLDP Configuration for the interface:

```
switch# show lldp configuration 1/1/1
LLDP Global Configuration
=====
LLDP Enabled                : Yes
LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : No
LLDP Port Configuration
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
1/1/1         Yes             Yes             Yes
```

Displaying LLDP Configuration for the management interface:

```
switch# show lldp configuration mgmt
LLDP Global Configuration
=====
LLDP Enabled                : Yes
LLDP Transmit Interval      : 30
LLDP Hold Time Multiplier   : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Timer Interval  : 2
LLDP Trap Enabled           : Yes
LLDP Port Configuration
=====
PORT          TX-ENABLED      RX-ENABLED      INTF-TRAP-ENABLED
-----
mgmt          Yes             Yes             Yes
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code> and <code>config -if</code>	Administrators or local user group members with execution rights for this command.

mac-notify traps

Syntax

```
mac-notify traps {aged | learned | moved | removed}  
no mac-notify traps {aged | learned | moved | removed}
```

Description

Configures a Layer 2 or VXLAN interface to generate SNMP trap notifications for up to four different types of MAC address related events on the trunk or access in physical or lag interfaces.

MAC notification trap addition to or removal from an interface can be in any combination, quantity, or order. The addition of existing configured traps or removal of non-configured traps will be accepted and ignored.



NOTE

The **mac-notify** feature must be enabled globally for any interface configurations to generate SNMP traps. Enabling **mac-notify traps** may impact the system performance on networks with a large number of mac-notify events.

The **no** form of this command removes the traps from the interface.

Parameter	Description
aged	Notifies when a MAC address aged out on the interface.
learned	Notifies when a MAC address is learned on the interface.
moved	Notifies when a MAC address moved from the interface.

Parameter	Description
removed	Notifies when a MAC address is removed from the interface.



NOTE

MAC notification cannot be configured on a Layer 3 (routing) interface. A Layer 2 interface that is changed to a Layer 3 interface through the `routing` command will discard any existing MAC notification configurations.

When MACs are learned on VXLAN tunnels or **port-access port-security** enabled ports, the move scenario is handled by the EVPN/port-access feature respectively. It performs the move by deleting the MAC from the old port and installing it on the new port. In this scenario, MAC trap notifications, if enabled, will reflect that by producing **removed** and **learned** notifications.

Usage

- MAC notify trap will not generate for static MACs.
- **vsx-sync** is not supported. You must enable the MAC notify traps explicitly on secondary to ensure the traps are generated.
- For EVPN MAC move between the following interfaces, the respective event types are produced (not always removed or learned)
 - Port to port: **moved**
 - Port to tunnel: **removed/learned**
 - Tunnel to port: **removed/learned**
 - Tunnel to Tunnel: **moved**

Examples



NOTE

MAC notification types and the associated events only apply to Layer 2 and VXLAN interfaces, hence routing might need to be disabled on the relevant interfaces.

Enable MAC notification traps within the SNMP module at a global level:

```
switch(config)# snmp-server trap
aaa-server-reachability-status  Enable SNMP trap for AAA server
reachability status
configuration-changes           Enable configuration changes traps
```

cpu-utilization	Enable high CPU utilization traps
link-status	Enable link status traps for all physical interfaces
mac-notify	Enable MAC table change notification traps
memory-utilization	Enable high memory utilization traps
module	Enable module event traps
port-security	Enable port-security violation traps. (Default: enable)
rmon-events	Enable RMON event traps
snmp	Enable snmp traps

For more information, see [snmp-server trap mac-notify](#).

Enabling the traps on an L2 interface:

```
switch(config)# interface 1/1/1
switch(config-if)# mac-notify traps learned
1/1/1 is not an L2 interface or tunnel
switch(config-if)# no routing
switch(config-if)# mac-notify traps learned removed
switch(config-if)# mac-notify traps moved
switch(config-if)# mac-notify traps aged
switch(config)# interface vxlan 1
switch(config-vxlan-if)# mac-notify traps learned removed
switch(config)# interface lag101
switch(config-if)# mac-notify traps removed
```

Disabling the `learned` and `removed` traps from the interface 1/1/1:

```
switch(config)# interface 1/1/1
switch(config-if)# no mac-notify traps learned removed
switch(config)# interface vxlan 1
switch(config-vxlan-if)# no mac-notify traps learned removed
```

Enable sending SNMP notifications for MAC table changes:

```
switch(config-vxlan-if)# mac-notify traps
  aged          Notify when a MAC address aged out on the interface
  learned       Notify when a MAC address was learned on the interface
  moved         Notify when a MAC address moved from the interface
  removed       Notify when a MAC address was removed from the interface
switch(config-vxlan-if)# mac-notify traps learned aged removed moved
```

Command History

Release	Modification
10.13.1000	Support for SNMP MAC notify traps on VXLAN tunnels.
10.10	Support for port access features with mac - notify added.
10.08	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command. Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

```
rmon alarm {enable | disable} {index | all}
```

Syntax

```
rmon alarm {enable | disable} {index <INDEX> | all}
no rmon alarm [enable | disable] [index <INDEX> | all]
```

Description

Enables and disables the RMON alarm and its index. RMON alarm is enabled by default.

Parameter	Description
enable	Enables the RMON alarm index
disable	Disables the RMON alarm index.

Parameter	Description
index <INDEX>	Specifies the RMON alarm index. Range: 1 to 20.
all	Specifies all the RMON alarms.

Examples

Enabling or disabling all the RMON alarm:

```
switch(config)# rmon alarm enable all
switch(config)# rmon alarm disable all
```

Enabling or disabling RMON alarm by index:

```
switch(config)# rmon alarm enable index 1
switch(config)# rmon alarm disable index 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

rmon alarm

Syntax

```
rmon alarm index <INDEX> snmp-oid <SNMP-OID> rising-threshold <RISING-THRESHOLD>
    falling-threshold <FALLING-THRESHOLD> [sample-interval <SAMPLE-INTERVAL>] [sample-type <ABSOLUTE|DELTA>]
no rmon alarm [index <INDEX>]
```

Description

Stores configuration entries in an alarm table that defines the sample interval, sample-type, and threshold parameters for an SNMP MIB object. Only the SNMP MIB objects that resolve to an ASN.1 primitive type of INTEGER (INTEGER, Integer32, Counter32, Counter64, Gauge32, or TimeTicks) will be monitored.

The **no** form of this command removes all RMON alarms and allows you to specify an index to remove a particular RMON alarm.

Parameter	Description
<code>index <INDEX></code>	Specifies the RMON alarm index. Range: 1 to 20.
<code>snmp-oid <SNMP-OID></code>	Specifies the SNMP MIB object to be monitored by RMON.
<code>rising-threshold <RISING-THRESHOLD></code>	Specifies the upper threshold value for the RMON alarm.
<code>falling-threshold <FALLING-THRESHOLD></code>	Specifies the falling threshold value for the RMON alarm. The falling threshold must be less than the rising threshold.
<code>sample-interval <SAMPLE-INTERVAL></code>	Sample interval in seconds. Default: 30.
<code>sample-type <ABSOLUTE DELTA></code>	Specifies the method of sampling of the SNMP MIB object. Default: Absolute.

Examples

Configuring RMON for the MIB object **ifOutErrors.15** with an index **1**, rising threshold of **2147483647** and falling threshold of **-2134** using **absolute** sampling for a sample interval of **100** seconds:

```
switch(config)# rmon alarm index 1 snmp-oid ifOutErrors.15 rising-threshold
2147483647
                falling-threshold -2134 sample-type absolute sample-interval 100
```

Removing RMON alarm with the index 5:

```
switch(config)# no rmon alarm index 5
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show configuration-changes trap

Syntax

```
show configuration-changes trap
```

Description

Shows the SNMP configuration changes trap settings.

Example

Showing the SNMP configuration changes trap:

```
switch# show configuration-changes trap
SNMP Configuration changes trap : Enabled
...
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mac-notify port

Syntax

```
show mac-notify [port <PORTS>]
```

Description

Displays the MAC notification configuration on a range of ports.

Parameter	Description
[port <PORTS>]	Specifies a port, range of ports, or list of ports.

Examples

Showing the MAC notification configuration on a range of ports:

```
switch(config)# show mac-notify port 1/1/1,1/1/3,1/1/5,lag101-lag104
MAC notification global Setting: Enabled
Port          Enabled Traps
-----
1/1/1          aged learned moved
1/1/3          --
1/1/5          moved
lag101         removed
lag102         --
lag103         --
lag104         aged learned moved removed
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mac-notify

Syntax

```
show mac-notify
```

Description

Displays whether the MAC notification feature in the SNMP module is enabled or not. It also displays the trap notification types configured on the Layer 2 ports in the system.

Examples

Showing the MAC notification configuration on all configured ports in the system:

```
switch# show mac-notify
MAC notification global setting : Enabled
Port          Enabled Traps
-----
1/1/1         aged learned moved
1/1/5         moved
lag101        removed
lag104        aged learned moved removed
...
...
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show rmon alarm

Syntax

```
show rmon alarm [index <INDEX>]
```

Description

Displays the RMON alarm configurations.

Parameter	Description
index <INDEX>	Specifies the RMON alarm index. Range: 1 to 20.

Examples

Showing all RMON alarm configurations:

```
switch# show rmon alarm
Index           : 1
Enabled         : true
Status          : valid
MIB object      : ifOutErrors.15
Sample type     : delta
Sampling interval : 6535 seconds
Rising threshold : 100
```

```
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11
Index : 3
Enabled : true
Status : invalid
MIB object : IF-MIB::ifDescr.19
Sample type : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
Falling threshold : 10
Last sampled value : 0
```

Showing RMON alarm with alarm index 1:

```
switch# show rmon alarm index 1
Index : 1
Enabled : true
Status : valid
MIB object : ifOutErrors.15
Sample type : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-06-21 05:58:11
```

Showing disabled RMON alarm information:

```
switch# show rmon alarm
Index : 1
Enabled : false
Status : valid
MIB object : ifOutErrors.15
Sample type : delta
Sampling interval : 6535 seconds
Rising threshold : 100
Falling threshold : 10
Last sampled value : 0
Last sample time : 2020-09-21 05:58:11
Index : 3
Enabled : false
Status : invalid
MIB object : IF-MIB::ifDescr.19
Sample type : absolute
Sampling interval : 10000 seconds
Rising threshold : 4000
```

```
Falling threshold : 10
Last sampled value : 0
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
-----------	-----------------	-----------

All platforms	config	Administrators or local user group members with execution rights for this command.
---------------	--------	--

show snmp agent-port

Syntax

```
show snmp agent-port
```

Description

Displays SNMP agent UDP port number.

Example

Displaying SNMP agent UDP port number:

```
switch# show snmp agent-port
SNMP agent port : 161
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp community

Syntax

```
show snmp community
```

Description

Displays a list of all configured SNMPv1/v2c communities.

Usage

When a user creates a custom community before enabling an SNMP agent, AOS-CX automatically removes the default **public** community from the system.

Example

Displaying a list of all configured SNMPv1/v2c communities:

```
switch#show snmp community
```

```
SNMP-COMMUNITIES
```

```
-----  
Community      Access-level  ACL Name    ACL Type    View  
-----  
private        ro           my_acl      ipv4         view1  
private        ro           my_acl      ipv6         none  
private2       rw           new_Acl     ipv6         view2  
private3       rw           none        none         none
```

When the switch is configured to use SNMPv3 only, the output of the **show snmp community** command displays the message **SNMP v1/v2c is disabled while snmpv3-only mode is configured**:

```
switch# show snmp community
```

```
-----  
-----  
Community      Access-level  ACL Name    ACL  
Type    View
```


SNMP v1/v2c is disabled while snmpv3-only mode is configured

Command History

Release	Modification
10.14	The output of this command now displays an error message when the switch is in SNMPv3 - only mode.
10.10	Output has been updated with SNMP view details. A View column is added to the command output.
10.08	Added ACL Type column to the command output.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show snmpv3 context

Syntax

```
show snmpv3 context
```

Description

Displays all configured SNMP contexts.

Examples

Displaying all configured SNMP contexts:

```
switch# show snmpv3 context
-----
name                                vrf                                community
-----
```

```

contextA                                default                                private

switch# show snmpv3 context
-----
Name          vrf          Community          type[Instance_id]
-----
A             default      public          vrf
switch#

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 engine-id

Syntax

```
show snmpv3 engine-id
```

Description

Displays the configured SNMPv3 snmp engine-id.

If the SNMPv3 engine-id is not configured, by default a unique engine-id is created by the switch using a combination of the enterprise OID value and the switch's mac address.

Example

Displaying the configured SNMPv3 engine-id:

```

switch# show snmpv3 engine-id
SNMP engine-id : 80:00:B8:5C:08:00:09:1d:de:a5

```


Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 security-level

Syntax

```
show snmpv3 security-level
```

Description

Displays the configured SNMPv3 security level.

Examples

Displaying the configured SNMPv3 security level:

```
switch# show snmpv3 security-level
SNMPv3 security-level : auth
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp system

Syntax

```
show snmp system
```

Description

Displays SNMP description, location, and contact information.

Example

Displaying SNMP description, location, and contact information:

```
switch# show snmp system
SNMP system information
-----
System description : Aggregation router
System location   : Main lab
System contact    : John Smith, Lab Admin
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp trap

Syntax

```
show snmp trap
```

Description

Displays all configured SNMP traps/informs receivers.

Example

Displaying all configured SNMP trap and informs receivers:

```
switch# show snmp trap
HOST          PORT  TYPE  VER  COMMUNITY/USER NAME  VRF      NOTIFICATION
TYPES
-----
-----
10.10.10.10    162   trap   v1   public              default   bgp
10.10.10.10    162   inform v2c  public              default   bgp, ospf,
fan, mstp
10.10.10.10    162   inform v3   name              default
```

Command History

Release	Modification
10.14	Updated the example output.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmpv3 users

Syntax

```
show snmpv3 users
```

Description

Displays all configured SNMPv3 users.

For more details on the user enabled status, see [snmpv3 security-level](#).

Example

Displaying all configured SNMPv3 users:

```
switch# show snmpv3 users
```

User	AuthMode	PrivMode	Status	Context	Access-level	View
name	md5	none	Enabled	context2 context1 context3	ro	view1
name2	none	none	Disabled	none	ro	view2
name3	none	none	Disabled	none	ro	none

Command History

Release	Modification
10.10	Output has been updated with SNMP view details. A View column is added to the command output.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp views

Syntax

```
show snmp views
```

Description

Displays the list of all the configured SNMP views.

Usage

The following table contains the status and its description of the configured SNMP views:

Status	Description
pending_validation	Default value that indicates SNMP view is yet to be validated.
operational	OID and mask validated.
invalid	Invalid OID/mask.
failed	Validation failed for reasons other than OID/mask.

Examples

Displaying the list of all the configured SNMP views:

```
switch# show snmp views
-----
SNMP MIB Views
-----
View      : new
OID Tree: sysUpTime.0
Mask      : ff
```

```
Type      : included
Status    : pending_validation
View      : admin
OID Tree  : ifIndex.1
Mask      : ff:a0
Type      : included
Status    : operational
View      : user
OID Tree  : sysb
Mask      : none
Type      : excluded
Status    : invalid
View      : admin
OID Tree  : .1.3.6.1.2.1.1
Mask      : none
Type      : excluded
Status    : operational
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show snmp vrf

Syntax

```
show snmp vrf
```

Description

Displays the VRF on which the SNMP agent service is running.

Example

Displaying SNMP services enabled on VRF:

```
switch#show snmp vrf
SNMP enabled VRF
-----
mgmt
default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

snmpv3 context

Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]
no snmpv3 context <NAME> [vrf <VRF-NAME>] [community <STRING>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The **no** form of this command removes the specified SNMP context.

Parameter	Description
	Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).

Parameter	Description
<code><NAME></code>	
<code>vrf <VRF-NAME></code>	Specifies the VRF associated with the context. Default: default.
<code>community <STRING></code>	Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config) # snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config) # snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config) # no snmpv3 context newContext vrf myVrf
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 engine-id

Syntax

```
snmpv3 engine-id <ENGINE-ID>
no snmpv3 engine-id <ENGINE-ID>
```

Description

Configures the SNMPv3 SNMP engine-id allowing an administrator to configure a unique SNMP engine-id for the switch. This engine-id is used by the NMS management tool to identify and distinguish multiple switches on the same network.

The **no** form of this command restores the default engine-id, created by the switch using a combination of the enterprise OID value and the switch's mac address.

Parameter	Description
<code><ENGINE-ID></code>	SNMPv3 SNMP engine - id in colon separated hexadecimal notation.

Examples

Configuring the SNMPv3 engine-id:

```
switch(config)#
switch(config)# snmpv3 engine-id
WORD SNMPv3 snmp engine-id in colon separated hexadecimal notation
switch(config)# snmpv3 engine-id 01:23:45:67:89:ab:cd:ef:01:23:45:67
```

Restoring the default SNMPv3 engine-id:

```
switch(config)# no snmpv3 engine-id
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 security-level

Syntax

```
snmpv3 security-level {auth | auth-privacy}  
no snmpv3 security-level {auth | auth-privacy}
```

Description

Configures the SNMPv3 security level. The security level determines which SNMPv3 users defined by the command `snmpv3 user` are able to connect.

The **no** form of this command changes the security level as follows:

- **no snmpv3 security-level auth:** Sets the security level to **auth-privacy**.
- **no snmpv3 security-level auth-privacy:** Sets the security level to no authentication or privacy, allowing any SNMP user to connect.

Parameter	Description
auth	SNMPv3 users that support authentication, or authentication and privacy are allowed.
auth-privacy	Only SNMPv3 users with both authentication and privacy are allowed. This is the highest level of SNMPv3 security. Default.

Examples

Setting the SNMPv3 security level to authentication and privacy:

```
switch(config)# snmpv3 security-level auth-privacy
```

Setting the SNMPv3 security level to authentication only:

```
switch(config)# snmpv3 security-level auth
```

Setting the SNMPv3 security level to no authentication and no privacy:

```
switch(config)# no snmpv3 security-level auth-privacy
```

Restoring the default SNMPv3 security level to authentication and privacy:

```
switch(config)# no snmpv3 security-level auth
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server agent-port

Syntax

```
snmp-server agent-port <PORT>  
no snmp-server agent-port [<PORT>]
```

Description

Sets the UDP port number that the SNMP master agent uses to communicate. UDP port 161 is the default port.

The **no** form of this command sets the SNMP master agent port to the default value.

Parameter	Description
<PORT>	Specifies the UDP port number that the SNMP master agent will use. Range: 1 to 65535. Default: 161.

Examples

Setting the SNMP master agent port to **2000**:

```
switch(config)# snmp-server agent-port 2000
```

Resetting the SNMP master agent port to the default value:

```
switch(config-schedule)# no snmp-server agent-port 2000
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server community view

Syntax

```
snmp-server community <STRING> [view <VIEW-NAME>]  
no snmp-server community <STRING> [view <VIEW-NAME>]
```

Description

Associates an SNMP MIB view with the SNMP community.

The **no** form of this command removes the associated SNMP MIB view from the SNMP community.

Parameter	Description
<STRING>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.
<VIEW-NAME>	Specifies the view name for the SNMP MIB view. Accepts a maximum of 32 characters.

Examples

Configuring the SNMPv1/SNMPv2c community:

```
switch(config)# snmp-server community my_community
switch(config-community)#
```

Adding SNMP MIB view to the SNMP community:

```
switch(config-community)# view name1
```

Removing SNMP MIB view from the SNMP community:

```
switch(config-community)# no view name1
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config config-community	Administrators or local user group members with execution rights for this command.

snmp-server community

Syntax

```
snmp-server community <STRING>
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is like a password that controls read/write access to the SNMP agent. A network management program must supply this name when attempting

to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The **no** form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value **public** is automatically defined.

Parameter	Description
<code><STRING></code>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Subcommands

```
access-level {ro | rw}
no access-level {ro | rw}
```

This subcommand changes the access level of the SNMP community. The default access level is read-only (**ro**).

The **no** form of this subcommand changes the access level of the community to default.

Parameter	Description
<code>ro</code>	Specifies Read - Only access with the SNMP community.
<code>rw</code>	Specifies Read - Write access with the SNMP community.

```
access-list {ip | ipv6} <ACL-NAME>
no access-list {ip | ipv6} <ACL-NAME>
```

This subcommand associates an ACL with the SNMP community. If an ACL is not associated with the SNMP community, the default access is allowed for all the hosts.

The **no** form of this subcommand removes association of the ACL with the SNMP community.

Parameter	Description
<code>ip</code>	Specifies the IPv4 ACL type.
<code>ipv6</code>	Specifies the IPv6 ACL type.
	Specifies the ACL name. It supports a maximum of 64 characters.

Parameter	Description
<ACL-NAME>	

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

Configuring the access level for the SMNP community to read-only:

```
switch(config-community)# access-level ro
```

Changing the access level of the SNMP community to default:

```
switch(config-community)# no access-level rw
```

Associating an IPv4 ACL named **my_acl** with the SMNP community:

```
switch(config-community)# access-list ip my_acl
```

Removing the associated IPv4 ACL named **my_acl** from the SNMP community:

```
switch(config-community)# no access-list ip my_acl
```

Configuration supported for SNMP ACL:

```
access-list ip ipv4_acl
  10 permit any 4.4.4.4 4.4.4.1
  20 permit any 3.3.3.3 3.3.3.1
access-list ipv6 ipv6_acl
  10 permit any 2001::2 2001::1
  20 permit any 3001::2 3001::1
snmp-server vrf default
snmp-server community my_comm_1
  access-list ip ipv4_acl
  access-list ipv6 ipv6_acl
```

Configuration not supported for SNMP ACL:

```
access-list ip ipv4_acl
  10 deny any 6.6.6.6 6.6.6.1
access-list ipv6 ipv6_acl
  10 deny any 6001::6 6000::1
```

```
snmp-server vrf default
snmp-server community my_comm_1
  access-list ip ipv4_acl
  access-list ipv6 ipv6_acl
```



NOTE

hitcounts for SNMP ACL will not be incremented.

Example: `show access-list hitcounts ip all` will not show the hit count of SNMP ACL.

Command History

Release	Modification
10.14	Replaced the ipv4 parameter with the ip parameter. The ipv4 parameter is deprecated.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config config- community	Administrators or local user group members with execution rights for this command.

snmp-server historical-counters-monitor

Syntax

```
snmp-server historical-counters-monitor
no snmp-server historical-counters-monitor
```

Description

Enables the Remote Network Monitoring agent (**rmond**) to start collecting historical interface statistics. The **no** form of this command stops the historical interface statistics collection.

Example

Enabling the `rmond` agent to start historical interface statistics collection:

```
switch(config)# snmp-server historical-counters-monitor
```

Disabling the `rmond` agent to stop historical interface statistics collection:

```
switch(config)# no snmp-server historical-counters-monitor
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (<code>></code>) or Manager (<code>#</code>)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

snmp-server host

Syntax

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION>
[community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>] [notification-type <NOTIFICATION-TYPE>]
```

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform
version v3] user <NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type
<NOTIFICATION-TYPE>]
no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform
version v3] user <NAME> [port <UDP-PORT>] [<VRF-NAME>] [notification-type
<NOTIFICATION-TYPE>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.



NOTE

Avoid configuring the same receiver for both SNMP traps and informs on the same UDP port for v1, v2, and v3.

Parameter	Description
<code><IPv4-ADDR></code>	Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100 .
<code><IPv6-ADDR></code>	Specifies the IP address of a trap receiver in IPv6 format (x:x::x:x).
<code>trap version <VERSION></code>	Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: v1 , v2c or v3 .
<code>inform version v2c</code>	Specifies the inform notification type for SNMPv2c.
<code>trap version v3</code>	Specifies the trap notification type for SNMPv3.
<code>user <NAME></code>	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.
<code>community <STRING></code>	Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public .
	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.

Parameter	Description
<code><UDP-PORT></code>	
<code><VRF-NAME></code>	Specifies the VRF on which the SNMP agent listens for incoming requests.
<code><notification-type></code>	Specifies the type of notification to be sent to the trap receiver. If no type is specified, all notifications are sent. The supported notification types are: • aaa - server • alarm • bgp • card • config • entity • fan • interface • lldp • loop - protect • mac - notify • mstp • mvrp • ospf • ospfv3 • port - security • power • power - ethernet • rmon • rpvst • stp • temperature • vrrp • vsf • vsx

Examples

```
switch(config) # snmp-server host 10.10.10.10 trap version v1
switch(config) # no snmp-server host 10.10.10.10 trap version v1
```

```

switch(config)# snmp-server host a:b::c:d trap version v1
switch(config)# no snmp-server host a:b::c:d trap version v1

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public

switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community
public

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d trap version v2c community public
port 5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public

switch(config)# snmp-server host a:b::c:d inform version v2c community
public
switch(config)# no snmp-server host a:b::c:d inform version v2c community
public

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public port 5000

```

```

switch(config)# snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default

switch(config)# no snmp-server host 10.10.10.10 inform version v2c
community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d inform version v2c community
public port 5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community
public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
port 2000

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port
2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin
port 2000

```

SNMP trap notification type examples:

```

switch(config)# snmp-server host 10.10.10.10 trap version v2c community
public notification-type bgp fan interface power entity
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community
public notification-type bgp
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp fan interface power-ethernet
switch(config)# no snmp-server host a:b::c:d inform version v3 user Admin
notification-type bgp interface
switch(config)# snmp-server host a:b::c:d inform version v3 user Admin
notification-type ?
    aaa-server      Sends AAA notifications.
    alarm           Sends Alarm notifications.
    bgp             Sends Border Gateway Protocol (BGP) state change

```

```

notifications.
  card          Sends Card notifications.
  config        Sends Configuration change notifications.
  entity        Sends Entity notifications.
  fan           Sends Fan notifications.
  interface     Sends Interface notifications.
  lldp          Sends Link Layer Discovery Protocol (LLDP) notifications.
  loop-protect  Sends Loop Protect notifications.
  mac-notify    Sends MAC Notify notifications.
  mstp          Sends Multiple Spanning Tree Protocol (MSTP)
notifications.
  mvrp          Sends Multiple VLAN Registration Protocol (MVRP)
notifications.
  ospf          Sends Open Shortest Path First (OSPFv2) notifications.
  ospfv3        Sends Open Shortest Path First version 3 (OSPFv3)
notifications.
  port-security Sends Port Security notifications.
  power         Sends Power notifications.
  power-ethernet Sends Power over Ethernet (PoE) notifications.
  rmon          Sends Remote Network Monitoring (RMON) notifications.
  rpvst         Sends Rapid Per VLAN Spanning Tree (RPVST) notifications.
  snmp          Sends Sends Simple Network Management Protocol (SNMP)
notifications.
  stp           Sends Spanning Tree Protocol (STP) notifications.
  temperature   Sends Temperature notifications.
  vrrp          Sends Virtual Router Redundancy Protocol (VRRP)
notifications.
  vsf           Sends Virtual Switching Framework (VSF) notifications.
  vsx           Sends Virtual System Extension (VSX) notifications.

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

snmp-server system-contact

Syntax

```
snmp-server system-contact <INFO>  
no snmp-server system-contact [<INFO>]
```

Description

Sets SNMP contact information.

The **no** form of this command removes the SNMP contact information.

Parameter	Description
<INFO>	Specifies SNMP contact information. Range: 1 to 128 printable ASCII characters, except for question mark (?).

Examples

Defines SNMP contact information to be **John Smith, Lab Admin**:

```
switch(config)# snmp-server system-contact John Smith, Lab Admin
```

Removes SNMP contact information:

```
switch(config)# no snmp-server system-contact
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server system-description

Syntax

```
snmp-server system-description <DESCRIPTION>
no snmp-server system-description
```

Description

Sets the SNMP system description.

The **no** form of this command removes the SNMP system description.

Parameter	Description
<DESCRIPTION>	Specifies the SNMP system description. Typical content to include would be the full name and version of the following: • Hardware type of the system • Software operating system • Networking software Range: 1 to 64 printable ASCII characters, except for the question mark (?).

Examples

Defines the SNMP system description to be **mainSwitch**:

```
switch(config)# snmp-server system-description mainSwitch
```

Removes the SNMP system description:

```
switch(config)# no snmp-server system-description mainSwitch
```


Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server system-location

Syntax

```
snmp-server system-location <INFO>  
no snmp-server system-location
```

Description

Sets the SNMP location information.

The **no** form of this command removes the SNMP location information.

Parameter	Description
<INFO>	Specifies the SNMP location information. Range: 1 to 128 printable ASCII characters, except for the question mark (?).

Examples

Defines the SNMP location information to be **Main Lab**:

```
switch(config)# snmp-server system-location Main Lab
```

Removes the SNMP location information:

```
switch(config)# no snmp-server system-location
```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
-----------	-----------------	-----------

All platforms	config	Administrators or local user group members with execution rights for this command.
---------------	--------	--

snmp-server trap-source interface vrf

Syntax

```
snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]  
no snmp-server trap-source {interface <IF-NAME> | <IPv4-Address> | <IPv6-Address>} [vrf <VRF-NAME>]
```

Description

Configures SNMP trap source interface or IP address for a VRF.

The **no** form of this command removes the SNMP **trap-source** configuration for a VRF.

Parameter	Description
-----------	-------------

<IF-NAME>	Specifies the source interface name. Interface name can be physical interface, loopback interface, LAG interface, or VLAN interface.
-----------	--

	Specifies the IPv4 address of source interface for the SNMP trap.
--	---

Parameter	Description
<code><IPv4-Address></code>	
<code><IPv6-Address></code>	Specifies the IPv6 address of source interface for the SNMP trap.
<code><VRF-NAME></code>	Specifies the name of a VRF associated to the source interface for the SNMP trap.

Examples

Configuring SNMP trap source interface for a VRF.

```
switch(config) # snmp-server trap-source interface 1/1/12 vrf sample
switch(config) # snmp-server trap-source interface loopback10 vrf sample
switch(config) # snmp-server trap-source interface vlan23 vrf sample
```

Configuring SNMP trap source IP address for a VRF.

```
switch(config) # snmp-server trap-source 10.0.0.1 vrf red
switch(config) # snmp-server trap-source 1001::1 vrf red
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server vrf

Syntax

```
snmp-server vrf <VRF-NAME>
no snmp-server vrf <VRF-NAME>
```

Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. 4100i, 6000, and 6100 only support default VRF. **The SNMP agent can listen on multiple VRFs.**

The **no** form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<VRF-NAME>	Specifies the name of a VRF.

Examples

Configuring the SNMP agent to listen on VRF **default**.

```
switch(config) # snmp-server vrf default
```

Configuring the SNMP agent to listen on VRF **mgmt**.

Stopping the SNMP agent from listening on VRF **default**.

```
switch(config) # no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user view

Syntax

```
snmpv3 user <USER-NAME> view <VIEW-NAME>
no snmpv3 user <USER-NAME> view <VIEW-NAME>
```

Description

Associates a user with an existing SNMP MIB view.

The **no** form of this command removes the associated user from the specified SNMP MIB view.

Parameter	Description
<USER-NAME>	Specifies the user name for the SNMP MIB view. Accepts a maximum of 32 characters.
<VIEWNAME>	Specifies the view name for the SNMP MIB view. Accepts a maximum of 32 characters.

Examples

Adding a user in the existing SNMP MIB view:

```
switch(config)# snmpv3 user nw-admin view my-nw-view
```

Removing the user from the SNMP MIB view:

```
switch(config)# no snmpv3 user nw-admin view my-nw-view
```

Attaching unconfigured or unknown SNMP view to an SNMPv3 user:

```
switch(config)# snmpv3 user nw-admin view myView
View myView is not configured.
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user

Syntax

```
snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
    [access-level ro|rw]
no snmpv3 user <NAME>
    [auth <AUTH-PROTO> auth-pass [{plaintext | ciphertext} <AUTH-PASS>]]
    [priv <PRIV-PROTO> priv-pass [{plaintext | ciphertext} <PRIV-PASS>]]
    [access-level ro|rw]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The SNMPv3 security level (set with command **snmpv3 security-level**) determines which users are allowed to authenticate.

The **no** form of this command removes the specified SNMPv3 user.



NOTE

When updating the authentication protocols and privacy protocols for the existing SNMPv3 users, you must also update the access level. Otherwise, the access level will be set to read-only.

Parameter	Description
<code><NAME></code>	Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).
<code>access - level</code>	Configures the access level for the SNMPv3 user: • ro: Allow read - only access for the SNMPv3 user • rw: Allow read - write access for the SNMPv3 user
<code>auth <AUTH-PROTO></code>	Sets the authentication protocol used to validate user logins. Supported protocols are md5 , sha , sha224 , sha256 , sha384 , and sha512 .
<code>auth-pass [{plaintext ciphertext} <AUTH-PASS>]</code>	Specifies the SNMPv3 user authentication password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for ciphertext is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches.  NOTE Authentication passwords that include special characters must be enclosed in single quotation marks ('). For example, 'auth - pwd20246!@#'.
<code>priv <PRIV-PROTO></code>	Sets the SNMPv3 privacy protocol (encryption method). Supported privacy protocols are aes , aes192 , aes256 , and des .
<code>priv-pass [{plaintext ciphertext} <PRIV-PASS>]</code>	Specifies the SNMPv3 user privacy encryption password. Range for plaintext is 8 to 32 printable ASCII characters, excluding space and question mark (?). Range for ciphertext is 1 to 256 printable ASCII characters. Ciphertext is used when copying user configuration settings between switches.  NOTE

Parameter

Description



Authentication passwords that include special characters must be enclosed in single quotation marks ('). For example, '**priv - pwd20246!@#**'.



NOTE

When the authentication password is not provided on the command line, plaintext authentication password prompting occurs upon pressing Enter, followed by privacy encryption protocol prompting, and finally plaintext encryption password prompting. The entered password characters are masked with asterisks.



NOTE

When the authentication type and password plus the privacy protocol (encryption method) are provided on the command line but the encryption password is not provided, plaintext encryption password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Defining SNMPv3 user **Admin1** using **sha** authentication and **des** privacy encryption with provided plaintext passwords:

```
switch(config)# snmpv3 user Admin1 auth sha auth-pass plaintext F82#450h
                  priv des priv-pass plaintext F82#4eva
```

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with provided authentication password and privacy encryption type but prompted encryption password:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass plaintext F82#450h
                  priv aes priv-pass
```

Enter the privacy encryption key: *****

Re-Enter the privacy encryption key: *****

Defining SNMPv3 user **Admin2** using **MD5** authentication and **AES** privacy encryption with plaintext password prompting and privacy encryption selection:

```
switch(config)# snmpv3 user Admin2 auth md5 auth-pass
```

Enter the authentication password: *****

Re-Enter the authentication password: *****

Configure the privacy protocol (y/n)? **y**

Enter the privacy protocol (aes/des)? **aes**


```
Enter the privacy encryption key: *****
Re-Enter the privacy encryption key: *****
```

Removing SNMPv3 user **Admin1**:

```
switch(config)# no snmpv3 user Admin1
```

Creating an SNMP user on switch 1 and then creating the same user on switch 2 by copying from the switch 1 configuration:

On switch 1, configure a user named **Admin3**, and then use the **show running-config** command to display switch configuration. Save a copy of the full **snmpv3 user** command (shown by **show running-config**). This saved command is used on switch 2.

```
switch1(config)# snmpv3 user Admin3 auth sha auth-pass plaintext F82#450h
                    priv des priv-pass plaintext F82#4eva
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version AOS-CX xx.xx.xx.xxxxxx
!
snmpv3 user Admin3 auth sha auth-pass ciphertext AQBaf2d...FJVcZ3o=
priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
ssh server vrf mgmt
!
interface mgmt
no shutdown
ip dhcp
vlan 1
```

On switch 2, execute the **snmpv3 user** command that you saved from switch 1 (as shown by **show running-config**). This creates the user on switch 2 with the same configuration.

```
switch2(config)# snmpv3 user Admin3 auth sha auth-pass ciphertext
AQBaf2d...FJVcZ3o=
                    priv des priv-pass ciphertext AQBah2p...2jfTFwQ=
```

The following command sets a read-write access level for an SNMPv3 user with the user name **user1**.

```
switch(config)# snmpv3 user user1 auth md5 auth-pass plaintext abc1234
access-level rw
```

Command History

Release	Modification
10.13	Following authentication protocols are supported: sha224 , sha256 , sha384 , and sha512 . Following privacy protocols are supported: aes192 and aes256 .
10.09	The access - level parameter was introduced.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server response-source

Syntax

```
snmp-server response-source {interface <IF-NAME> | <IPv4-ADDRESS> | <IPv6-ADDRESS>} [vrf <VRF-NAME>]  
no snmp-server response-source {interface <IF-NAME> | <IPv4-ADDRESS> | <IPv6-ADDRESS>} [vrf <VRF-NAME>]
```

Description

Configures the source interface or IP address for sending SNMP responses. Each SNMP can independently have its own unique response source IP address.

The **no** form of this command removes the source interface name or IP address for sending SNMP responses.



NOTE

- It is recommended to use the loopback interface or ip address of the loopback interface as the response source. If a device does not support a loopback interface, then configure SVI interface or SVI IP address as the response source.
- The active gateway IP address cannot be configured as the response source.
- It is recommended to limit the maximum number of response source to five.
- The interface used for the response source should be in the up state. If the interface is down, the default source IP will be used.
- The use of **udp6** is mandatory for IPv6 SNMP operations. For example, you can use the following syntax: **snmpwalk -v2c -c public -m ALL udp6: [2100::2] .1.3.6.1.2.1.1.**

Parameter	Description
<code>interface <IF-NAME></code>	Specifies the source interface name. The interface can be a physical interface, loopback interface, or VLAN interface.
<code><IPv4-ADDRESS></code>	Specifies the IPv4 address of the source interface for the SNMP response.
<code><IPv6-ADDRESS></code>	Specifies the IPv6 address of the source interface for the SNMP response.
<code>vrf <VRF-NAME></code>	Specifies the VRF name associated to the source interface for the SNMP response.

Examples

Configuring a response source for the interface **1/1/12**:

```
switch(config) # snmp-server response-source interface 1/1/12 vrf red
```

Configuring a response source for interface **loopback10**:

```
switch(config) # snmp-server response-source interface loopback10 vrf red
```

Configuring a response source for the IPv4 address **10.0.0.1**:

```
switch(config) # snmp-server response-source 10.0.0.1 vrf sample
```

Configuring a response source for the IPv6 address **2001::1**:

```
switch(config) # snmp-server response-source 2001::1 vrf default
```

Command History

Release	Modification
10.13	Added support for IPv6 address.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server snmpv3-only

Syntax

```
snmp-server snmpv3-only  
no snmp-server snmpv3-only
```

Description

Accepts SNMPv3 messages only, SNMPv1 and SNMPv2c will be disabled. By default SNMPv1, SNMPv2c and SNMPv3 will all be enabled.

The **no** form of this command restores the default setting and reenables SNMPv1 and SNMPv2c .

Examples

Configuring SNMPv3 messages only, and disabling SNMPv1 and SNMPv2c:

```
switch(config)# snmp-server snmpv3-only
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap

Syntax

```
snmp-server trap {cpu-utilization | memory-utilization | rmon-events}  
no snmp-server trap {cpu-utilization | memory-utilization | rmon-events}
```

Description

Enables the SNMP traps. The SNMP traps are enabled by default.

The **no** form of this command disables the SNMP traps.

Parameter	Description
cpu-utilization	Enables the CPU utilization traps.
memory-utilization	Enables the memory utilization traps.
rmon-events	Enables the RMON event traps.

Examples

Enabling the SNMP traps:

```
switch(config)# snmp-server trap cpu-utilization
switch(config)# snmp-server trap memory-utilization
switch(config)# snmp-server trap rmon-events
```

Disabling the SNMP traps:

```
switch(config)# no snmp-server trap cpu-utilization
switch(config)# no snmp-server trap memory-utilization
switch(config)# no snmp-server trap rmon-events
```

Displaying the SNMP trap configuration:

```
switch(config)# show running-config all | inc snmp
snmp-server trap rmon-events
snmp-server trap cpu-utilization
snmp-server trap memory-utilization
```

Displaying CPU and Memory usage:

```
switch(config)# show system
Hostname           : XXXX
System Description : XX.10.07.0001CI
System Contact     :
System Location    :
Vendor             : Aruba
Product Name       : JLXXXX XXXX Base Chassis/3xFT/18xFans/Cbl Mgr/X462
Bundle
Chassis Serial Nbr : SG6Z009068
Base MAC Address   : f40343-806400
AOS-CX Version     : XX.10.07.0001CI
Time Zone          : UTC
Up Time            : 8 minutes
CPU Util (%)       : 1
Memory Usage (%)   : 10
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap aaa-server-reachability-status

Syntax

```
snmp-server trap aaa-server-reachability-status
no snmp-server trap aaa-server-reachability-status
```

Description

Enables the SNMP trap for AAA server status. When enabled, traps are sent whenever AAA server (RADIUS, TACACS) status changes from reachable to unreachable and vice versa.

The **no** form of this command disables sending SNMP trap for AAA server status.

Examples

Enabling the SNMP trap for AAA server status:

```
switch(config)# snmp-server trap aaa-server-reachability-status
```

Disabling the SNMP trap for AAA server status:

```
switch(config)# no snmp-server trap aaa-server-reachability-status
```

Command History

Release	Modification
10.10	Command introduced on 4100i, 6000, 6100, 8100, 8320, 8325, 8360, 8400, 9300, and 10000 switches.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap configuration-changes

Syntax

```
snmp-server trap configuration-changes
no snmp-server trap configuration-changes
```

Description

Enables sending SNMP traps whenever the configuration changes. Configuration trap generation is disabled by default.

The **no** form of this command disables sending SNMP traps for configuration changes.

Parameter	Description
configuration-changes	Specifies SNMP traps for configuration changes.

Examples

Enabling the SNMP traps for configuration changes:

```
switch(config)# snmp-server trap configuration-changes
```

Disabling the SNMP traps for configuration changes:

```
switch(config)# no snmp-server trap configuration-changes
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap mac-notify

Syntax

```
snmp-server trap mac-notify
no snmp-server trap mac-notify
```

Description

Enables the MAC notification traps within the SNMP module at a global level. When enabled, traps are sent for interfaces that are configured for MAC notification events.

The **no** form of this command disables sending MAC notification traps at a global level. When disabled, existing **mac-notify** interface configuration is preserved but MAC notification events on configured interfaces will not cause SNMP traps to be transmitted.

Examples

Enabling the SNMP MAC notification feature in the system globally:

```
switch(config) # snmp-server trap mac-notify
```

Disabling the SNMP MAC notification feature in the system globally:

```
switch(config) # no snmp-server trap mac-notify
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server trap port-security

Syntax

```
snmp-server trap port-security  
no snmp-server trap port-security
```

Description

Enables SNMP port-security violation traps on the system. Port-security violation traps are enabled by default.

The **no** form of this command disables the SNMP port-security violation traps on the system.

Parameter	Description
port-security	Specifies SNMP traps for port - security.

Examples

Enabling the SNMP port-security violation traps on the system:

```
switch(config) # snmp-server trap port-security
```

Disabling the SNMP port-security violation traps on the system:

```
switch(config) # no snmp-server trap port-security
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

snmp-server trap snmp

Syntax

```
snmp-server trap snmp {authentication | coldstart | warmstart} [vrf  
<VRF_NAME>]  
no snmp-server trap snmp {authentication | coldstart | warmstart} [vrf  
<VRF_NAME>]
```

Description

Enables SNMPv2 MIB traps. The SNMPv2 traps are disabled by default.

The **no** form of this command disables the SNMPv2 MIB traps.

SNMPv2 MIB supports the following traps:

- authentication:** Authentication trap is sent when the SNMP server receives a protocol message that is not properly authenticated.
- coldstart:** A coldstart trap is sent when the switch reboots.
- warmstart:** A warmstart trap is sent when there is a user intervention to enable or disable the SNMP service on the switch.



NOTE

SNMPv2 Authentication traps do not support source IP configuration.

Parameter	Description
authentication	Enables the authentication traps.
coldstart	Enables the coldstart traps.

Parameter	Description
warmstart	Enables the warmstart traps.
<VRF_NAME>	Specifies the VRF name. Enables the SNMPv2 traps for a VRF.

Examples

Enabling all SNMPv2 traps:

```
switch(config) # snmp-server trap snmp
```

Enabling only SNMPv2 authentication traps:

```
switch(config) # snmp-server trap snmp authentication
```

Disabling all SNMP traps:

```
switch(config) # no snmp-server trap snmp
```

Command History

Release	Modification
10.10	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

```
snmp-server trap vsx
```

Syntax

```
snmp-server trap vsx
no snmp-server trap vsx
```

Description

Enables sending the SNMP traps for VSX related events. VSX trap generation is disabled by default.

The **no** form of this command disables sending the SNMP traps for VSX related events.

The trap support is available for the following VSX events:

- ISL up and down
- KA up and down
- MCLAG up and down

Parameter	Description
vsx	Specifies SNMP traps for VSX events.

Examples

Enabling the VSX traps:

```
switch(config)# snmp-server trap vsx
switch(config)# show vsx configuration trap
SNMP traps : Enabled
```

Disabling the VSX traps:

```
switch(config)# no snmp-server trap vsx
switch(config)# show vsx configuration trap
SNMP traps : Disabled
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server view

Syntax

```
snmp-server view <VIEWNAME>
                <OID_TREE> [<MASK>] <included/excluded>
no snmp-server view <VIEWNAME>
                <OID_TREE> [<MASK>] <included/excluded>
```

Description

Configures an SNMP MIB view.

The **no** form of this command removes the specified SNMP MIB view.

Parameter	Description
<VIEWNAME>	Specifies the name of the SNMP MIB view. Supports up to a maximum of 32 characters.
<OID_TREE>	Specifies the OID tree to be included or excluded in SNMP MIB view.
<MASK>	Specifies the OID mask value. The values must be in hexadecimal character separated with : (colon).
	Specifies the OID tree that is included in or excluded from the SNMP MIB view.

Parameter	Description
<code><included/excluded></code>	

Usage

You can configure a maximum of 50 SNMP MIB views. The following VTY message is displayed when the configuration exceeds the maximum SNMP MIB views:

```
switch(config)# snmp-server view name51 1.3.6.1.2.1.1 fe:00 included
Configuration failed: Maximum allowed views are configured.
```

Examples

Configuring the SNMP MIB views:

```
switch(config)# snmp-server view name1 .1.3.6.1.2.1.2.2.1.1.1 FF:A0 included
switch(config)# snmp-server view name2 IF-MIB::ifindex included
switch(config)# snmp-server view name4 1.3.6.1.2.1.1 fe:00 included
```

Removing an SNMP MIB view:

```
switch(config)# no snmp-server view name4 1.3.6.1.2.1.1 fe:00 included
```

Command History

Release	Modification
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Support and Other Resources

Subtopics

Accessing HPE Aruba Networking Support
Accessing Updates
Warranty Information
Regulatory Information
Documentation Feedback

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html
AOS - CX Switch Software Documentation Portal	https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	https://www.hpe.com/psnow/doc/a50011948enw

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS - CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release : https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End - of - Life information	https://networkingsupport.hpe.com/end - of - life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at [**https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**](https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts)

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see [**https://www.arubanetworks.com/company/about-us/environmental-citizenship/**](https://www.arubanetworks.com/company/about-us/environmental-citizenship/).

Documentation Feedback

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See Also